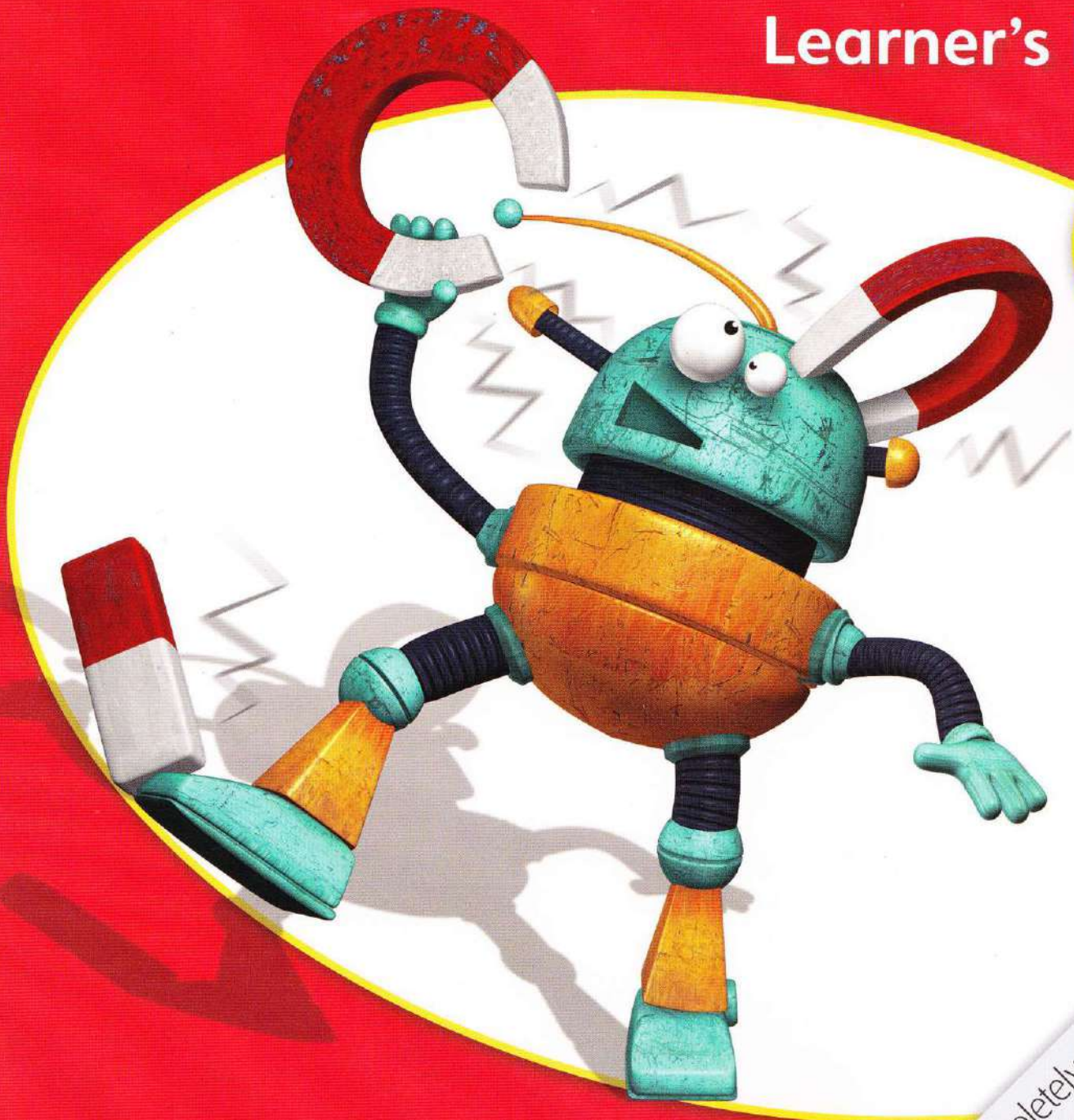
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CAMBRIDGE PRIMARY Science

Learner's Book 3



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Science

Learner's Book



Jon Board and Alan Cross

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Introduction

The *Cambridge Primary Science* series has been developed to match the Cambridge International Examinations Primary Science curriculum framework. It is a fun, flexible and easy-to-use course that gives both learners and teachers the support they need. In keeping with the aims of the curriculum itself, it encourages learners to be actively engaged with the content, and to develop enquiry skills as well as subject knowledge.

This Learner's Book for Stage 3 covers all the content in Stage 3 of the curriculum framework. The topics are covered in the order in which they are presented in the curriculum for easy navigation, but can be taught in any order that is appropriate to you.

Throughout the book you will find ideas for practical activities, which will help learners to develop their scientific enquiry skills as well as introduce them to the thrill of scientific discovery.

The 'Talk about it!' question in each topic can be used as a starting point for classroom discussion, encouraging learners to use scientific vocabulary and develop their understanding.

'Check your progress' questions at the end of each unit can be used to assess learners' understanding. Learners who will be taking the Cambridge Primary Progression test for Stage 3 will find these questions useful preparation.

We strongly advise you to use the Teacher's Resource for Stage 3, ISBN 978-1-107-61150-4, alongside this book. This resource contains extensive guidance on all the topics, ideas for classroom activities, and guidance notes on all the activities presented in this Learner's Book. You will also find a large collection of worksheets, and answers to all the questions from the Stage 3 products.

Also available is the Activity Book for Stage 3, ISBN 978-1-107-61150-4. This book offers a variety of exercises to help learners consolidate understanding, practise vocabulary, apply knowledge to new situations and develop enquiry skills. Learners can complete the exercises in class or be given them as homework.

We hope you enjoy using this series.

With best wishes,
the Cambridge Primary Science team.

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1 Looking after plants

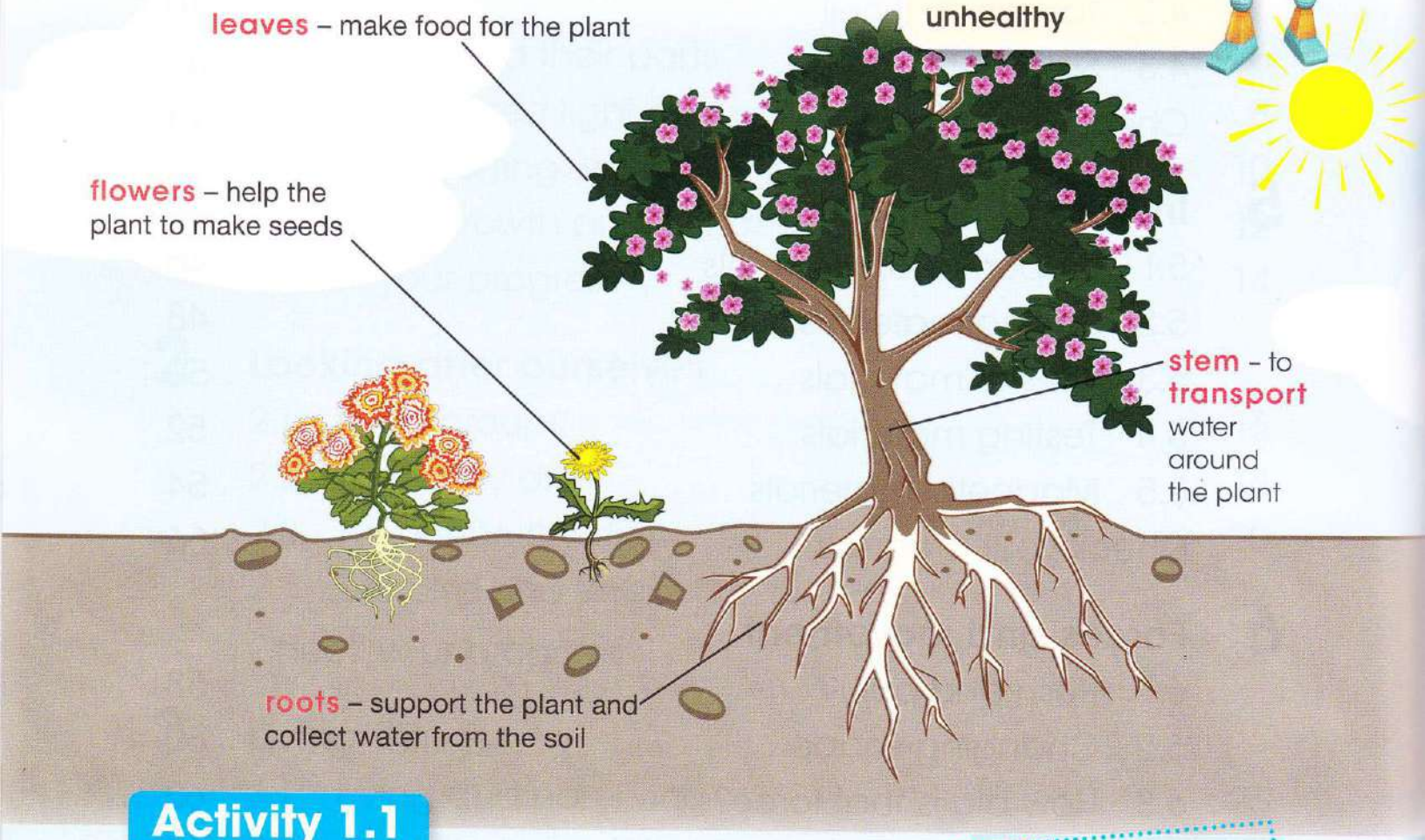
1.1 Plants and their parts

Plants can look very different.
Most have four main parts.

Words to learn

leaves flowers
stem transport
roots healthy
unhealthy





Activity 1.1

Make a model plant

Make a plant like this one.

Make labels for the stem, roots, leaves and flower.
Stick them to your plant.



You will need:
string • straws
coloured paper or card
sticky tape • glue



Healthy leaves.

Healthy plants have healthy roots, stems and leaves.



A plant with healthy roots.

A plant with **unhealthy** roots, stems and leaves will not grow well.



Unhealthy roots with root knot disease.



Unhealthy leaves with rust disease.

Questions

1. List **four** things that most plants have.
2. Describe how plants are different from each other.

Talk about it!

Why don't plants have flowers all year round?

What you have learnt

- Most plants have roots, a stem, leaves and flowers.
- Healthy plants have healthy roots, stem and leaves.
- A plant with unhealthy roots, stem and leaves will not grow well.

1.2 Plants need light and water



Sunil has an unhealthy plant.

What does a plant need to stay healthy and grow?

Words to learn

predict	investigation
fair test	wilt
explain	

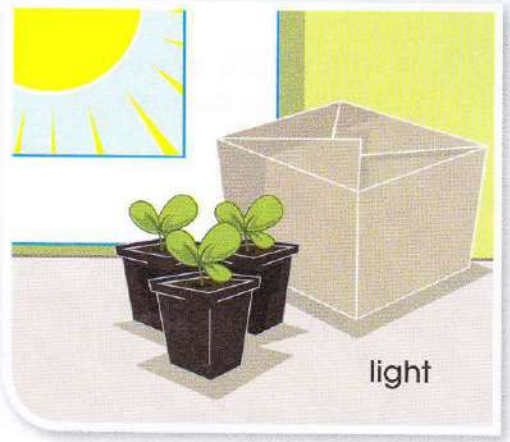


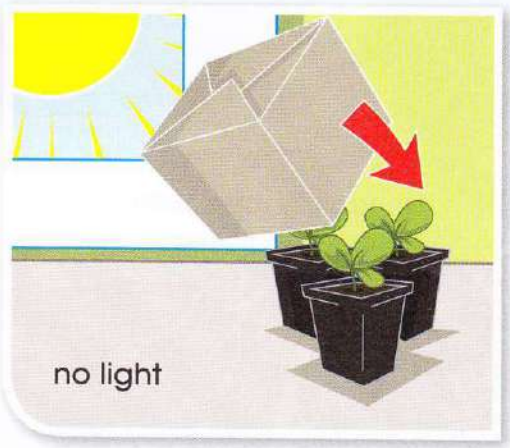
Activity 1.2a

Do plants need light to grow?

You will need:
six young plants • watering can • box to cover three plants

Look at the pictures to see what to do.





Water all the plants every day.
What do you **predict** will happen to the leaves, stems and roots of the plants?
In this **investigation** both groups of plants are given water.
What else do you need to keep the same to make this a **fair test**?

If plants do not get enough light and water the roots, stem or leaves, become unhealthy. The plant will **wilt.**

Light and water both help the plant to make food.

Activity 1.2b

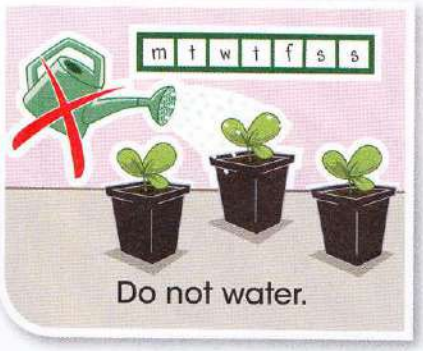
You will need:
nine young plants • watering can

Do plants need water to grow?

Look at the pictures to see what to do.







What do you predict will happen to the plants' leaves, stems and roots in each case?

What do plants need to grow? How do you know?

Explain what you have found out.

Questions

1. What does a plant need to grow and stay healthy?
2. Explain what happens to the stem and leaves when you put a plant in a dark place.
3. Would a plant grow if it was given juice instead of water? Draw a picture to show how you could investigate this.

The surface of the Moon.



Talk about it!
Could plants ever grow on the Moon?

What you have learnt

- All plants need light to grow and keep their stems and leaves healthy.
- All plants need some water to grow and keep their stems and leaves healthy.

1.3 Transporting water

The roots and the stem

Helene has a plant that looks unhealthy. Look at the roots.

Words to learn
absorb





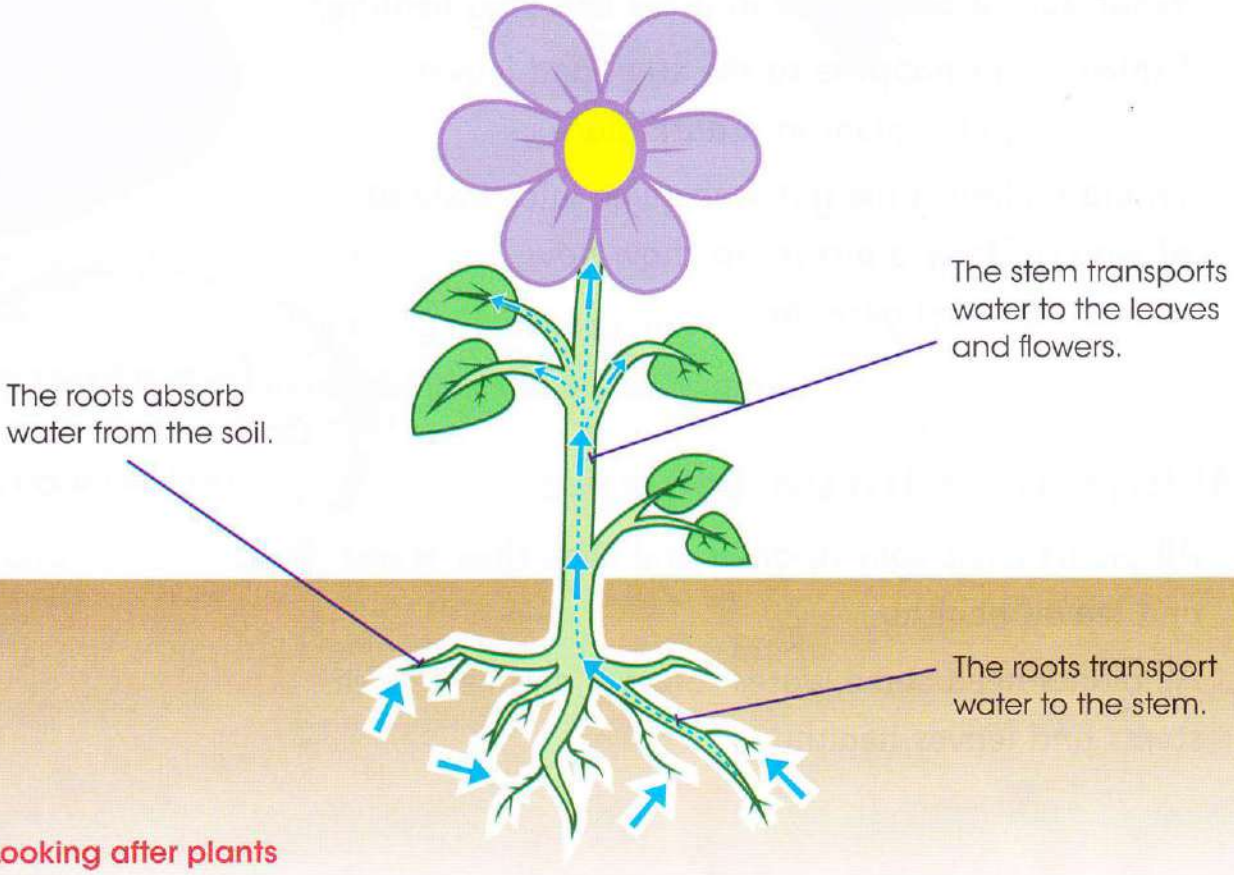
This pot is too small for the roots.



The roots are really close together and cannot **absorb** water well.



Putting the plant in a bigger pot will allow the roots to spread out.



Activity 1.3

Making plants go blue

You will need:
a white flower • some celery • a container of water
blue food colouring • a plastic knife

A plant stem can transport water to the leaves and flowers.
Predict what would happen if the water was blue.



After one day, cut the celery and look inside.
Can you explain what has happened to the celery and the flower?

Questions

1. Why do plants sometimes need to be moved to a bigger pot?
2. How do plants get water to their leaves?
3. What would happen to a plant with no roots? Why?

Talk about it!
What happens to flowers when they are cut and put into water?

What you have learnt

- Roots absorb water from the soil.
- Roots transport water to the stem.
- The stem transports water around the plant.

1.4 Plant growth and temperature

Activity 1.4

You will need:
two similar bean plants • a thermometer • a ruler

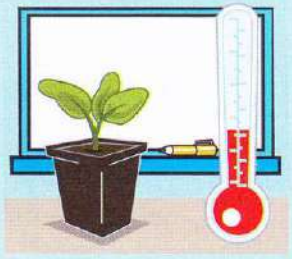
Words to learn

thermometer
temperature
results
bar chart
conclusion



At which temperatures do plants grow best?

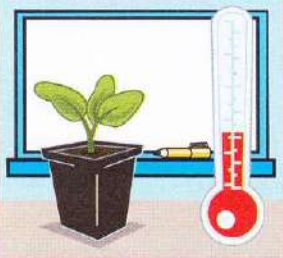
Set up an investigation like one of the pictures below.



Put one plant in the classroom.



Put one plant outside in cold weather.



Put one plant in the classroom.



Put one plant outside in warm weather.

Predict which plant will grow the best. Use a ruler to measure the height of the plants and a **thermometer** to measure the **temperature** every few days. Record your **results** in a table.

Day	Cold/hot plant		Warm plant	
	Temperature in °C	Height in cm	Temperature in °C	Height in cm
1				

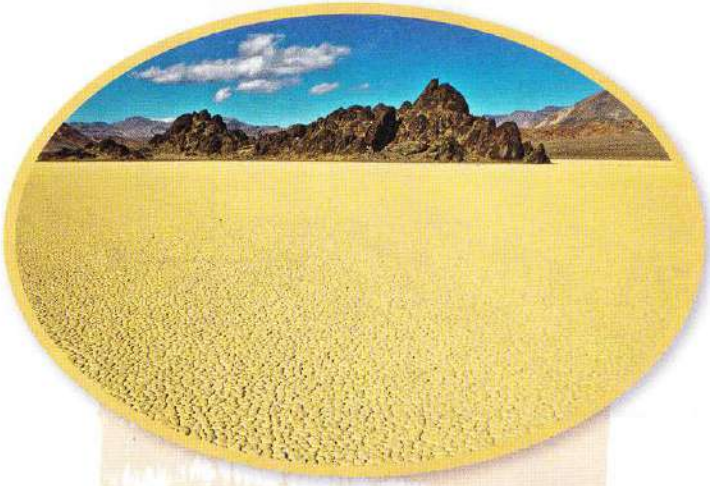
Draw a **bar chart** to show the height of the plants on the last day. What is your **conclusion**? At what temperatures do plants grow best?

Think about the place you live. Is there enough warmth for plants to grow?



This is a rainforest. Many plants find it easy to grow here. It is warm but not too hot and there is plenty of water.

When is the warmest time of year? When is the coldest? Can plants grow all year?



At over 56 °C, Death Valley in America is one of the hottest places on Earth.



At –89 °C, Antarctica is the coldest place on Earth.

Questions

1. Where do plants grow best? Cold places, hot places or warm places?
2. What happens when a plant gets too hot?
3. How is plant growth affected by temperature?

Talk about it!

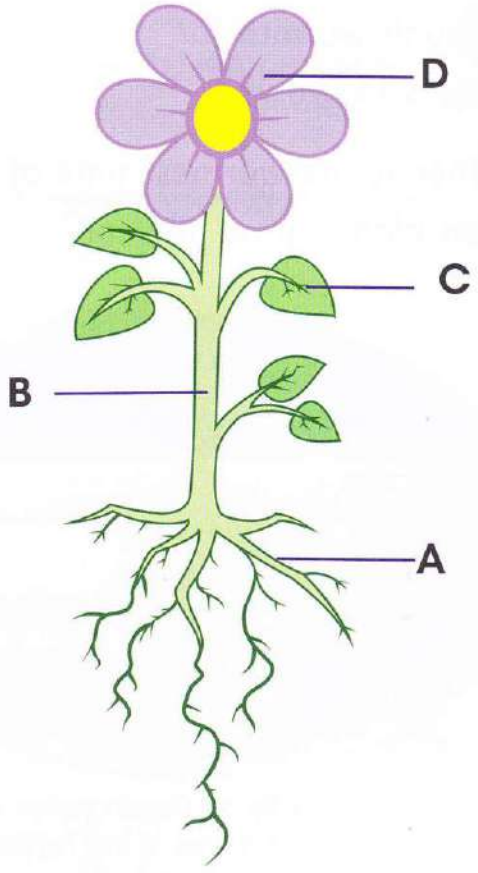
Why are there no plants in Death Valley or Antarctica?

What you have learnt

- Plants grow more slowly when it is cold.
- Plants grow more quickly when it is warm.
- Plants die when they get too cold or too hot.

1 Check your progress

1 Here is a flowering plant.



Copy and complete these sentences.

A is the _____ .
B is the _____ .
C is a _____ .
D is the _____ .

2 Copy and complete the sentences using the missing words.

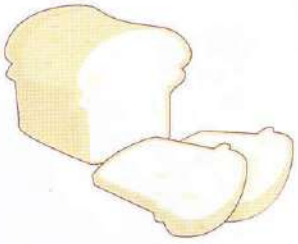
You may use each word more than once.

roots flowers stem leaves water

Plants have _____ that are under the ground. These hold the plant up and also absorb _____ . The water is transported to the _____ and then to the _____ and the _____ .

The _____ make food for the plant. The _____ help the plant to make seeds.

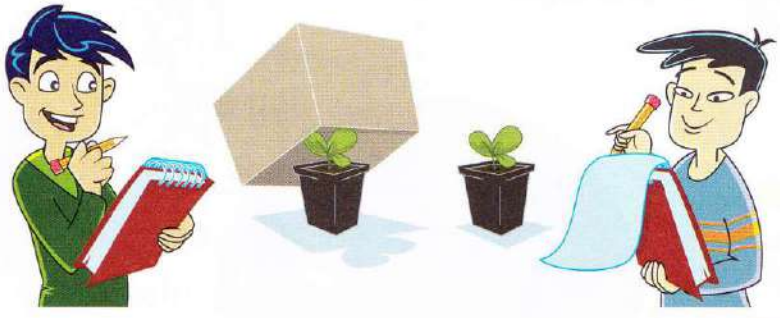
3 Which **two** of these does a plant need for it to to grow?

water 	light 
food 	plant pot 

Copy and complete this sentence.

A plant needs _____ and _____ to grow.

4 Luiz and Cheng have planned an investigation.



a What are they investigating?

b What do they need to keep the same to make the test fair?

c Which plant will grow the best?

2 Looking after ourselves

2.1 Food groups

Foods can be put into groups.

Words to learn

dairy	fruit
vegetables	fat
carbohydrate	energy
protein	meat
fish	





You need **dairy** foods for strong bones and teeth.





You need **fruit** and **vegetables** to be healthy.

Food groups





You should not eat too much **fat** and sugar.





You need **carbohydrate** for **energy**.





You need **protein** (such as **meat** and **fish**) for growth.

Activity 2.1

You will need:
real or model food • food packaging or pictures of food • food group name cards

Sorting food into food groups

Look at some foods and put them into food groups.
Use this book to help you.
The labels on the packets will help.

Bimla wants to be healthy. What should she eat?



Questions

1 Which food group should you eat least? Why?
2 Which food group helps you to grow?
3 What type of food group would be good to eat as a healthy snack?

Talk about it!
Which food groups should you eat most?

What you have learnt

- Food can be put into groups: carbohydrate, fruit and vegetables, protein, dairy, fat.

2.2 A healthy diet



Words to learn

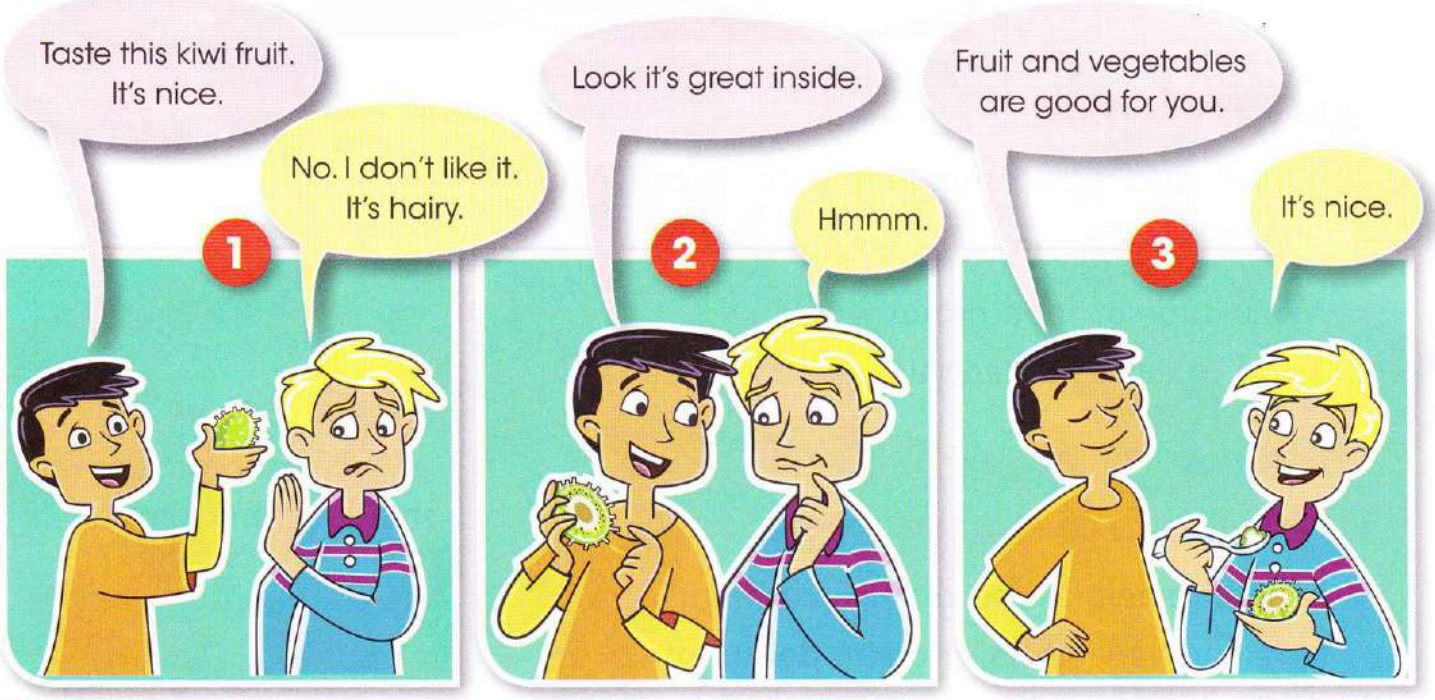
diet	nutrition
headache	dehydrated



A healthy **diet** needs to be varied. A balance of food from different groups gives your body the **nutrition** it needs.

This food triangle shows how much of each food group you should eat.

Fruit and vegetables keep us healthy. Sam will only eat bananas and carrots. He does not like any others.



Activity 2.2

You will need:
some fruit and vegetables you can taste
disposable spoons

Tasting fruit and vegetables

Talk about the taste of some fruit and vegetables.
Which ones will you like? Make a prediction then taste them.
Draw a table like this for your results.

Fruit/Vegetable	Prediction	Result	
	Will you like it?	How did it taste?	Did you like it?
melon	no	very sweet	yes



When running long distances it is easy to become dehydrated.

Water is part of a healthy diet. Not drinking enough will make you tired and give you a **headache**. This is called being **dehydrated**.

We can only live without water for a few days.

Questions

1. Why do we need to eat lots of fruit and vegetables?
2. What makes people dehydrated?

Talk about it!
Why was it good that Sam tasted the kiwi fruit?

What you have learnt

- A healthy diet gives your body the nutrition it needs.
- A food triangle shows how much of each food group you should eat.
- Fruit and vegetables are a very important part of a healthy diet.
- We can only live without water for a few days.

2.3 An unhealthy diet

Activity 2.3a

You will need:
some drinks and their labels

Words to learn
record salt
discuss



Which drinks have the most sugar?



Predict which drinks are healthy and which are unhealthy. Read the labels to find out how much sugar is in each drink. **Record** your results in a table.

Activity 2.3b

You will need:
a drink with lots of sugar • water
egg shells • 2 plastic cups

What does sugar do to teeth?

The material that makes up egg shells is similar to the material that makes up your teeth. Look at the pictures to see what to do. Predict what you think will happen to the egg shells.



Think about what you found out. Explain what too much sugar can do to your teeth.

Eating lots of fat, sugar and **salt** makes your diet unhealthy. Fat and salt are bad for your heart.

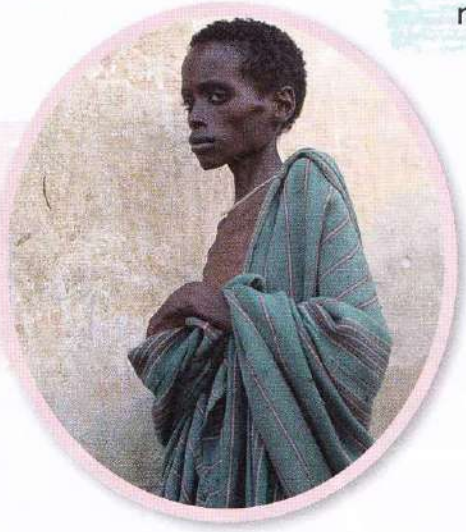


Sugar is bad for your teeth. Brushing your teeth after breakfast and before going to bed will help.



Eating too much food is bad for you. Your body can store food as fat. This can make you overweight.

Some people do not have enough food to eat. Being very thin is not healthy.



Challenge Is fruit juice good for your teeth?
How could you find out?
Discuss this with your friends.

Questions

Copy and complete these sentences.

1 Too much sugar is bad for your _____ .

2 Too much fat and salt is bad for your _____ .

Talk about it!
Why do we like to eat unhealthy food?

What you have learnt

- Too much fat, salt and sugar in your diet will make you unhealthy.

2.4 Exercise and sleep

To be healthy you need to look after your body in different ways.

You need a healthy diet, **exercise** and sleep.

Who is doing exercise in this picture?

Word to learn
exercise



Exercise is lots of movement.

Playing, running and walking are all good exercise.

Exercise makes your heart, bones and muscles strong.

Sleep is also important for keeping healthy.

Most school children need 10–12 hours of sleep.



Activity 2.4

You will need:
lots of space • stopwatch

What happens when we exercise?

Look at the pictures to see what to do.



Challenge Predict what will happen if you exercise for longer.



Even astronauts have to exercise to keep healthy.

Questions

1. Which of these is exercise: running, jumping, reading, sleeping, skipping, playing football?
2. Copy and complete the sentence. Exercise is good for your _____, _____ and _____.

Talk about it!
What is a good time to go to bed?

What you have learnt

- Exercise and sleep will help to keep you healthy.

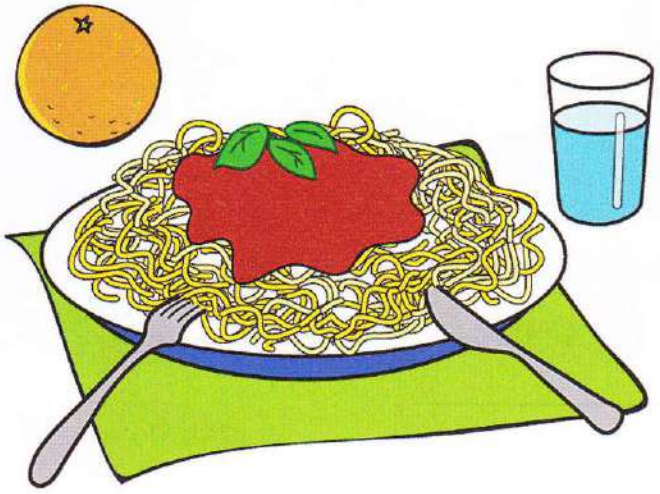
2 Check your progress

1 Match the food group with the description.

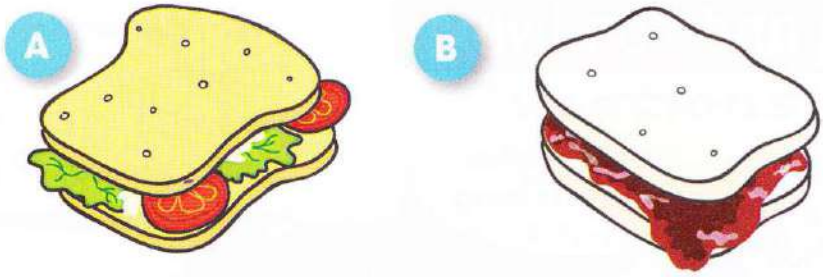
1	dairy
2	carbohydrate
3	fruit and vegetables
4	fat and sugar
5	protein

A	do not eat too much of this
B	gives you energy
C	helps you to grow
D	keeps your body healthy
E	keeps your bones and teeth strong

2 Here is a healthy meal. Explain why it is healthy.



3 Which sandwich is more healthy? Why?



4 These drinks have different amounts of sugar per 100ml.



Draw a table like this to show how much sugar is in each drink.

Drink	Sugar per 100ml in g

Draw a bar chart to present your results.

3 Living things

3.1 Living and non-living

Words to learn

breathe	young
grow	excrete
life processes	sort





Do you agree with what the learners say?

Which things are alive?

Which things in these pictures are alive?










Things that are alive:

- need air to **breathe**
- need water and food
- can move
- have senses
- can produce **young**
- **grow**
- produce waste products (**excrete**).

These are the seven **life processes**.

Questions

1. List the **seven** life processes.
2. A horse can run, eat, drink and see. Is it alive?
3. A toy kite moves and flies in the air. Is it alive?

Activity 3.1

Living or non-living?

You will need:
a large sheet of paper • pens
a selection of pictures of things that are living and non-living

With a partner, **sort** pictures of things into two groups:

- living things
- non-living things.

Use the seven life processes to help you decide where to put each picture.
Were there any that you found difficult to place? If so, explain why.

Talk about it!
How do you know if something is living?

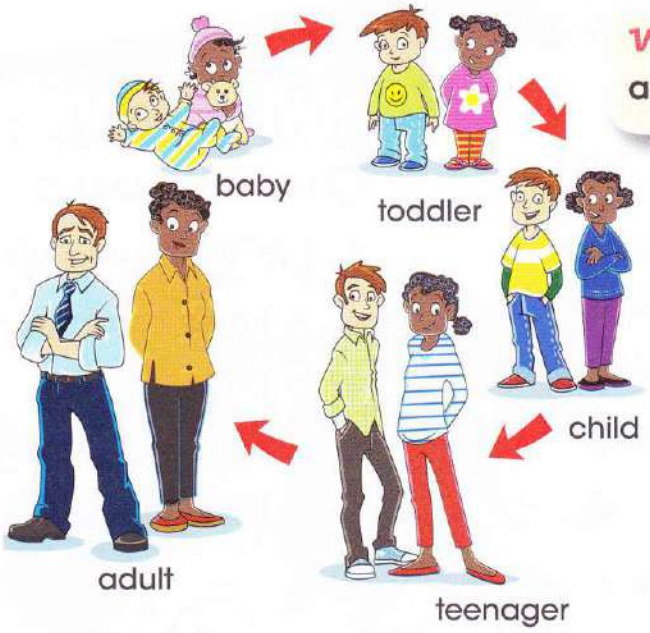
What you have learnt

- All living things need air to breathe, need water and food, can move, have senses, can produce young, grow and produce waste products.

3.2 Growth and nutrition

Growth

Look at these pictures. Where are you now in the cycle?



Words to learn
adult feed



All living things have young. The young grow. An **adult** is fully grown.

Name the young for each of these animals.


bear


cat


frog

Plants grow in stages.

1



A seed.

2



The seed begins to grow.

3



A young plant develops.

4



The young plant becomes a fully grown flowering plant.

Nutrition

Animal food comes from plants and other animals. Some animals feed their young.

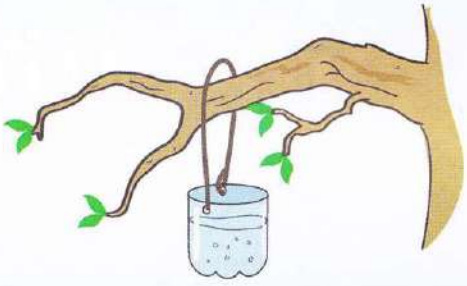


Plants make their own food. Plants use the light from the Sun and carbon dioxide from the air to make sugars. They make sugar in the green parts of the plant and release oxygen as a waste product.

Activity 3.2

Make a bird feeder

You will need:
a plastic container • string • a hole punch • fat • bird seed



Punch two holes near the top of the plastic container.
Melt the fat (get an adult to help you).
When the fat has cooled a little, stir bird seed into it.
Carefully pour the mixture into the plastic container (do not fill it to the top).
When the mixture is cold, thread the string through the holes and hang your bird feeder somewhere outside.

Questions

1. What are the stages of growth for a human being?
2. How do plants make their food?
3. Where does human food come from?

Talk about it!
What food is eaten by animals that you know?

What you have learnt

- Animals and plants grow.
- Plants make food by using sunlight to make sugars.
- Animals' food comes from plants and other animals.

3.3 Movement and reproduction

Movement

All animals and plants move.

Words to learn
reproduce offspring





Plants move slightly towards the light.

They can also spread across areas as their roots systems grow.

Their seeds can also move.

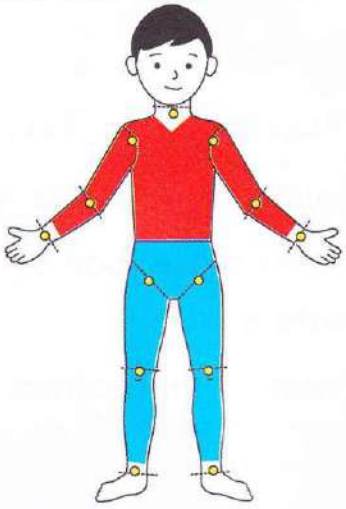
Activity 3.3

Make a model of the moving parts of your body

You will need:
a template of parts of the body • split pins

Cut out the parts of the body from the template.

Use split pins to join the parts together.



Reproduction

Living things **reproduce** so that new animals and plants can grow to become adults. Plants make seeds. Animals and birds lay eggs or have babies.



A bird brings material to make a nest.

When animals reproduce they often make a special home for the young or the eggs. Some animals carry their young.



Two polar bear cubs with their mother at their den.

Some animals do not prepare a home. The **offspring** have to look after themselves.

Fish swim off into the sea without parents.

Caterpillars are left to look after themselves.



A baby kangaroo in its mother's pouch.

Questions

1. Name an animal and the shelter it makes for its eggs or young.
2. Why are young caterpillars left to look after themselves?

Talk about it!

Why do some parents make a safe place for their eggs or young?

What you have learnt

- All animals and plants move and all living things reproduce.

3.4 Sorting humans

Look at other people. They are all **similar** to you but also **different**. Our eyes are different colours.

Words to learn

similar	different
fingerprint	identify
data	tally



These children all have similar bodies but there are differences. What two things are similar? What two things are different?



loop



arch



whorl

People have different shapes on their **fingerprint**.

Look at your fingers. Use a magnifying glass if possible. **Identify** which type of fingerprint you have.

Activity 3.4

You will need:
paper • clipboard • pencil

Collecting data

Collect **data** about your classmates.
Make a chart like this to help you.

Name	Hair colour	Eye colour	Height in m
Sunita	black	brown	1.21

Collect the names of your classmates and data about their hair, eyes and height.
Then **tally** the data on a tally chart like this one.

Hair colour	Brown	Black	Blonde
tally	𝍸 II	II	III
total	7	2	3

Questions

1. What are the common eye colours in human beings?
2. In what ways are we the same as other people?
3. How are we different from one another?

Talk about it!
What differences are we born with?

What you have learnt

- People are similar in some ways but different in others.
- Fingerprints are all different.
- People have different hair colour, eye colour and heights.

3.5 Sorting living things

Scientists who study living things need to be able to **group** them.

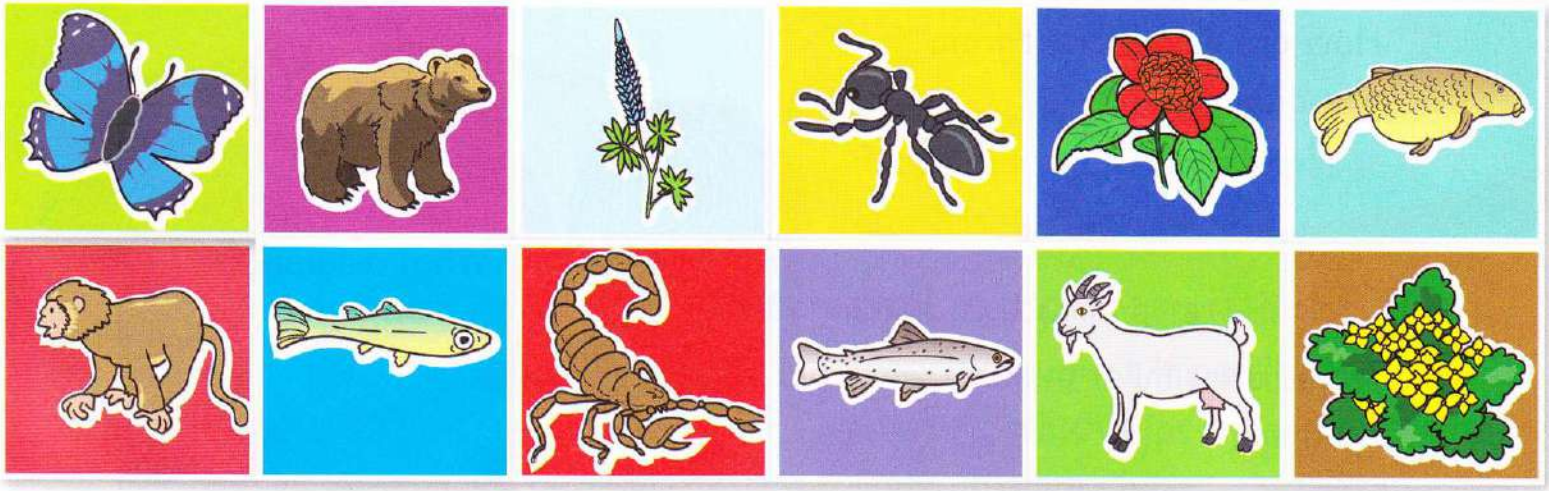
Words to learn
group question





Help the zookeeper sort the animals into the correct group.

Look at these pictures.
Talk to a friend about how you could sort these living things.



Describe the things that are similar about the things in your groups.

Juma has collected these small animals. Suggest groups he can put them in.

Suggest a **question** that he could use to sort the animals.



Activity 3.5

Sorting leaves

You will need:
a selection of different leaves • hoops
labels for groups

Sort your leaves into groups.
Put each group into a hoop.
Label each group.
Could you have sorted the leaves in a different way?

Questions

1. How many groups of animals and plants do you know? What are they?
2. Why do scientists need to group animals?
3. Why might you group a lion and a wolf together?

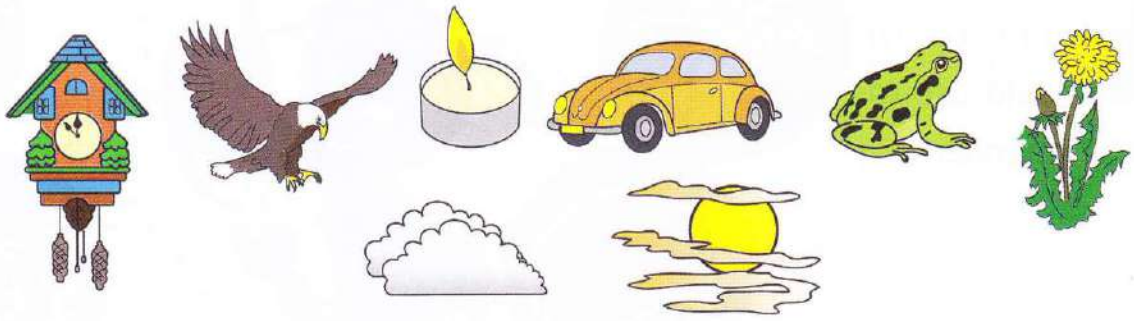
Talk about it!
What groups do animals and plants around your home belong to?

What you have learnt

- Living things can be put into groups.
- The groups have something in common.

3 Check your progress

1 Magda's book contains pictures of these things. Help her to say which are alive and which are not alive.



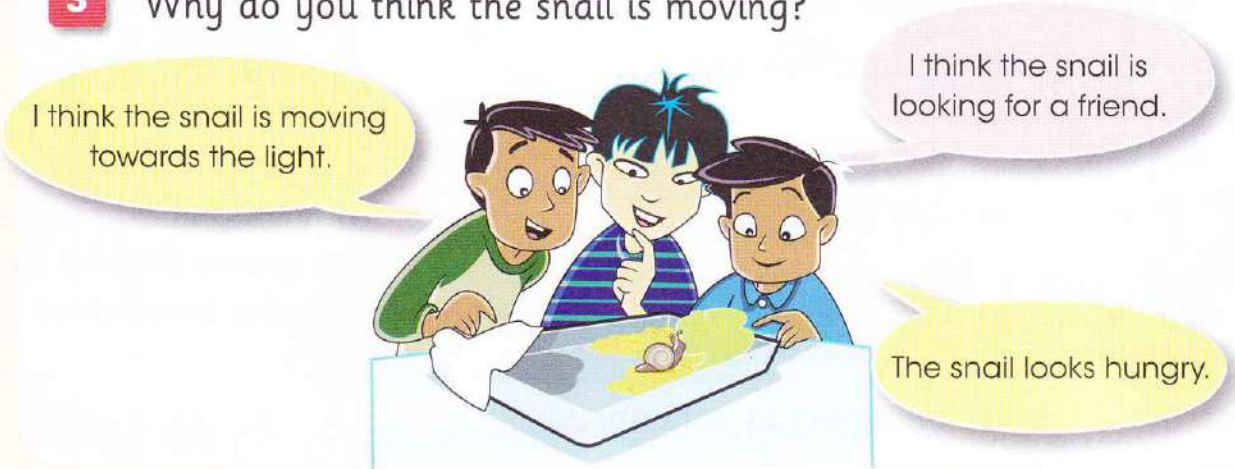
2 Here are some life stages of living things.

a	_____	→	tadpole	→	frog
b	egg	→	_____	→	hen
c	_____	→	child	→	adult
d	seed	→	seedling	→	_____
e	baby	→	foal	→	_____

Copy and complete each line. Use these words to help you.

horse frog spawn chick plant baby

3 Why do you think the snail is moving?



4 Salif has found this table about the number of young produced by some animals.

Animal	Number of young produced
rabbit	8
birds	6
human	1 or 2
fish	1000s of eggs
spiders	100s of eggs

a Which animals can have the most young?

b Why do some animals have many young?

5 Here is a plant.



For this plant, which questions are 'true' and which questions are 'false'?

a Does the plant have green leaves?

b Does the plant have a flower?

c Is the plant a tree?

d Does it have berries?

4 Our five senses

4.1 Hearing and touch

A sense is a way of finding out about the world around you.
The senses are touch, sight, hearing, smell and taste.

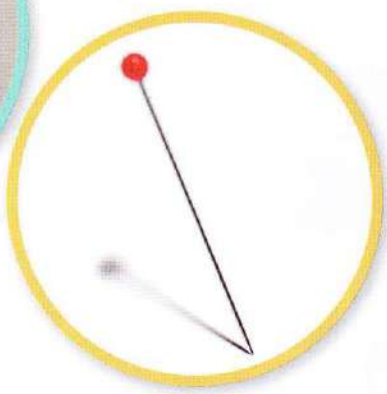
Hearing

Your ears are hearing all the time.





Which of these makes the loudest sound? Which is the quietest?





Activity 4.1a

Pointing at sounds

You will need:
material to act as a blindfold • something to make a sound (small bell or two spoons)

Blindfold a friend.
Make a sound.
Ask your friend to point to the source of the sound.
What can you conclude from their action?



Touch

We have all hurt ourselves.
All of your skin can feel things touching it: hot and cold, rough and smooth, dry and wet.



Activity 4.1b

Touch test

You will need:
wet and dry tissues • ice cubes • sandpaper
wood • plastic • metal • a blindfold

Blindfold a partner.
Give your partner each item in turn (the order does not matter).
Ask them to touch each item and describe how it feels.
Can they tell you what it is?



Questions

1 How many senses do you have?
2 **a** Give an example of a very quiet sound.
b Give an example of a very loud sound.
3 Where on your body is your sense of touch?

Talk about it!
If you were blind, how important would your other senses be?

What you have learnt

- Humans have five senses: hearing, touch, taste, smell and sight.
- The senses work together to help humans find out about the world around them.
- You use your ears to hear and your skin to touch.

4.2 Taste and smell

A **tongue** has taste buds which sense tastes like salt and **sweet**, **bitter** and **sour**.

Taste works with smell to stop us from eating food that would make us ill.

If something smells bad it generally tastes bad too.

If your nose is blocked you cannot taste very well.

Some animals have small noses.
Others have big noses.

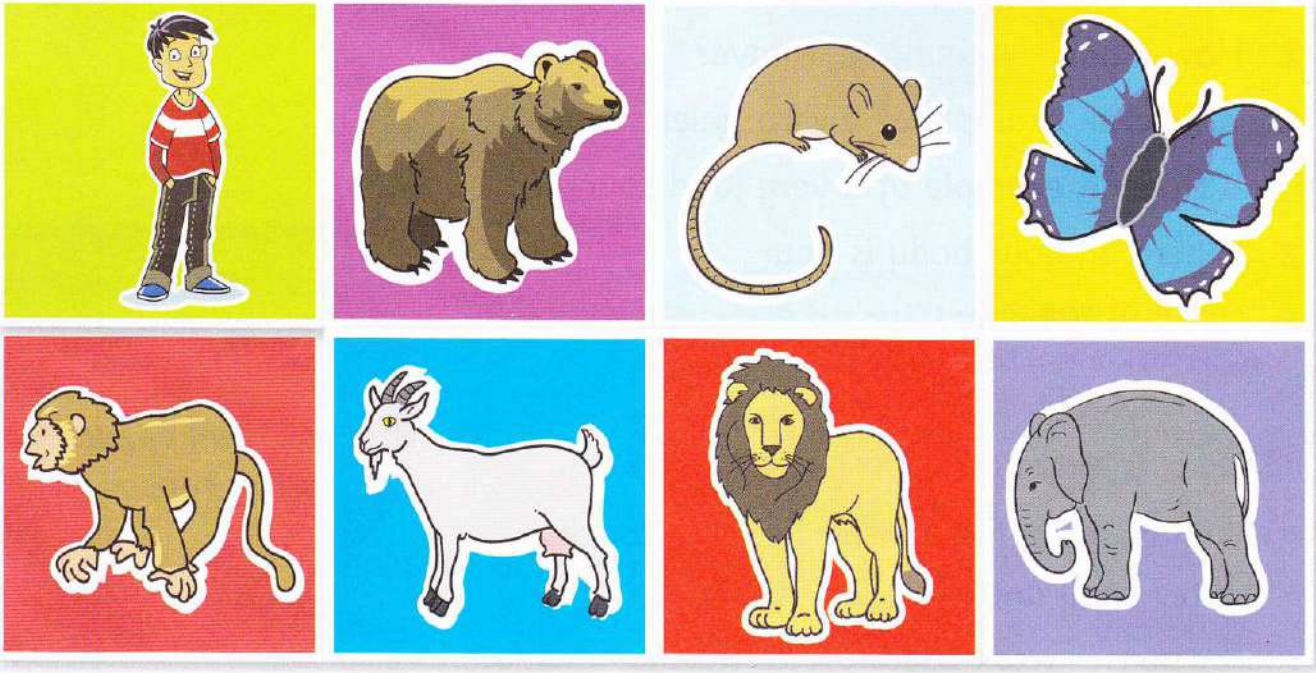
Words to learn

tongue	sweet
bitter	sour





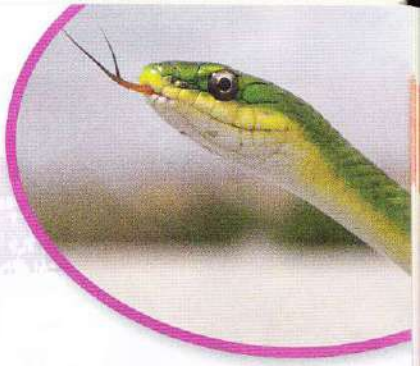
Which of these animals do you think can smell things the best?





Bears have a very good sense of smell. Bears can smell food far away.

Snakes smell with their tongues.



Animals which find their food on the ground have a good sense of smell.

Activity 4.2

Do you like or dislike this food?

You will need:
clean food samples on separate plates
clean spoons • plates • blindfold

Blindfold a friend.

Ask them to taste each sample of food and say whether they like it or not.

Repeat the test but, this time, ask your friend to hold their nose so that they cannot smell the food. Does this affect how they taste the food?



Questions

1. What part of our body do we use to taste things?
2. What part of our body do we use to smell things?
3. Why do we need to smell and taste foods?

What you have learnt

- The senses of taste and smell often work together.

Talk about it!

Why do some animals have a very good sense of smell?

4.3 Sight





What colour eyes do you and your friends have?

Word to learn
eyesight

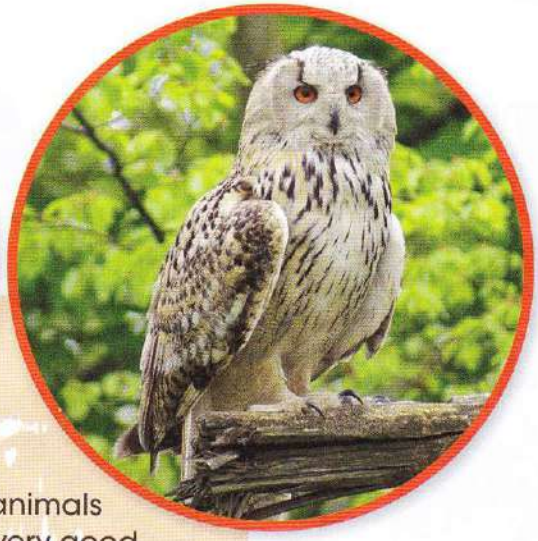




Your eyes are fantastic. They can see things close to you and much further away.

Some animals have very good **eyesight** and some animals do not.

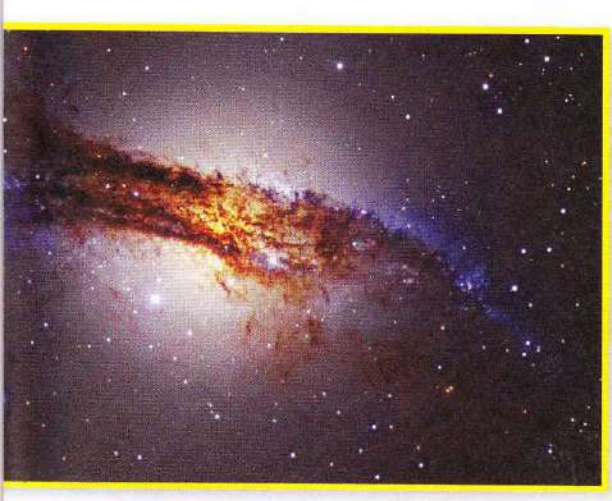




These animals use their very good eyesight for hunting prey.



This mole does not have very good eyesight. Instead, the mole uses his nose to find his way around.



Centaurus A, a galaxy that can be seen with naked eyes. It is around 12 million light years from Earth.

With our eyes we can see what is happening in the world. You can test your eyes.

Activity 4.3

Make and use an eye test

You will need:
a large piece of paper • a metre ruler
something to cover one eye

Choose a set of letters or numbers.
Print these in different sizes on your piece of paper.
Fix the paper 2 or 3 metres away.
Cover one eye while you read the letters or numbers out to check that you can see them clearly.
Now check your other eye.
Try your test with other people.
How far down the chart could you read the letters?

Was the result the same for both eyes?
Did other people have the same result?



Questions

1 Why do some animals have good eyesight?
2 Give an example of where a human needs to see things:
   a close up
   b at some distance away.

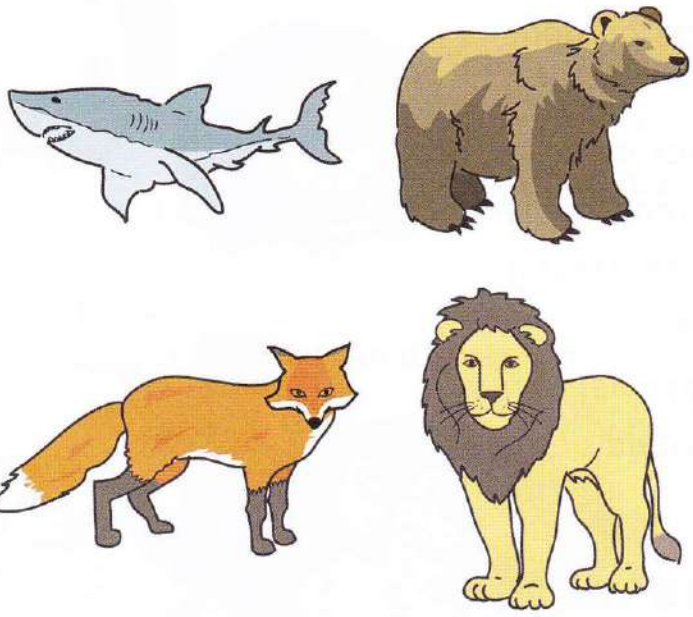
Talk about it!
How do animals who do not have good eyesight find out about their world?

What you have learnt

- Some animals have very good eyesight.
- Human eyes are a variety of colours.
- The human eye can see things that are very close and things that are some distance away.

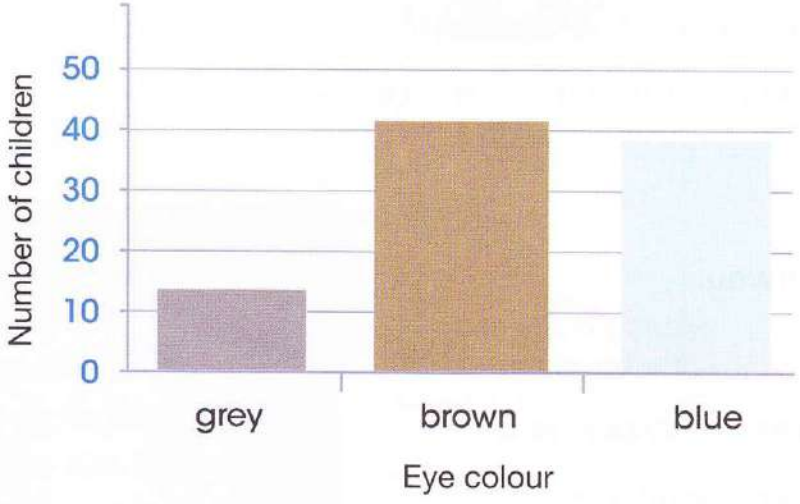
4 Check your progress

1 These animals have a good sense of smell.



Give two reasons why they need a good sense of smell.

2 Alex has recorded the eye colour of the children in his school on this graph.



a Which is the most common eye colour?

b Which is the least common eye colour?

3 Amal tested his friends to see which things they like and dislike.

Name	Food	Like	Dislike
Bill	apple	✓	
	biscuit	✓	
Jaffa	apple	✓	
	biscuit	✓	
Pat	apple		✓
	biscuit	✓	
Rava	apple	✓	
	biscuit	✓	

a Which food is most popular?

b Why do children like apples and biscuits?

4 **a** Describe how to carry out a taste test. Include details of the equipment you would use.

b How could you investigate whether the sense of taste and the sense of smell are linked?

5 Investigating materials

5.1 Properties of materials

Words to learn

property flexible
rigid waterproof
absorbent



A **property** describes what a material is like.





Metal is strong but paper is weak.





Rubber is **flexible** (it can be bent) but stone is **rigid** (it keeps its shape).





Plastic is **waterproof** (water cannot get through it) but cotton is **absorbent** (it soaks up liquid).

Activity 5.1

A materials hunt

You will need:
some materials

Find some materials.
Look carefully and feel the materials.
What properties do they have?
Draw a table like this and write down the properties of the materials.

Material	Properties
paper	weak, flexible, smooth, absorbent

Describe the properties of a material to a partner.
Can they guess which material it is?

Questions

1 List as many different materials as you can.

2 Which material is:

a silver, shiny and strong

b transparent, smooth, rigid and weak?



Yuri Gagarin wore the first space suit in space in 1961.

What you have learnt

- There are many different materials.
- Materials have many different properties.

Talk about it!
Which materials would you use to make a space suit?

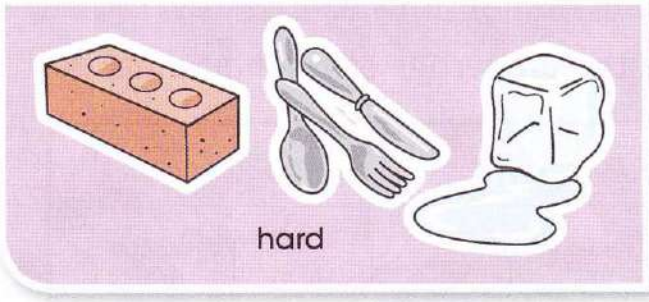
5.2 Sorting materials

A material can be put in a group. These materials are sorted into **hard** and **soft** groups.

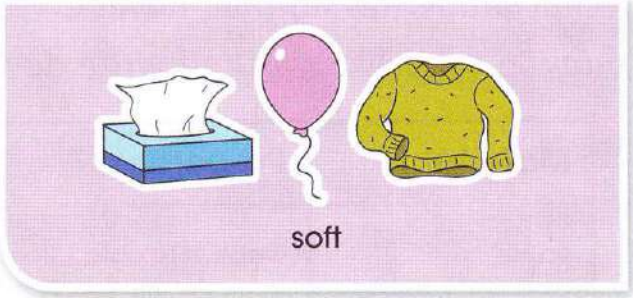
Words to learn

hard	soft
dull	shiny



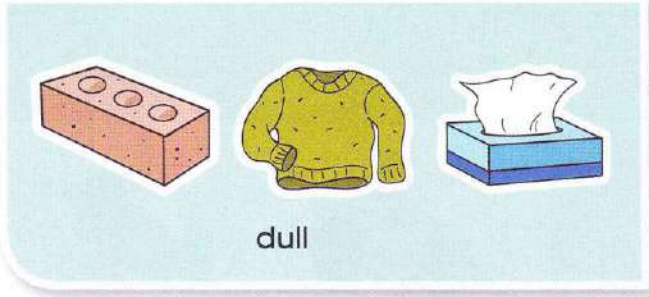


hard



soft

The same objects can also be sorted into **dull** and **shiny** groups.



dull



shiny

Activity 5.2

You will need:
some objects each made from a single material

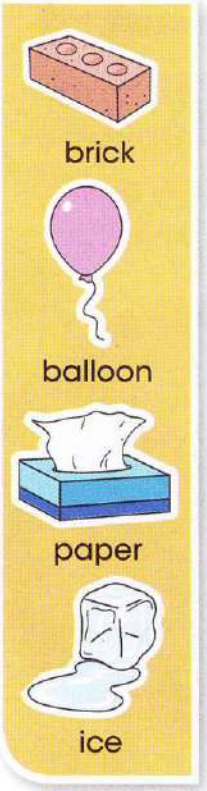
Sorting materials

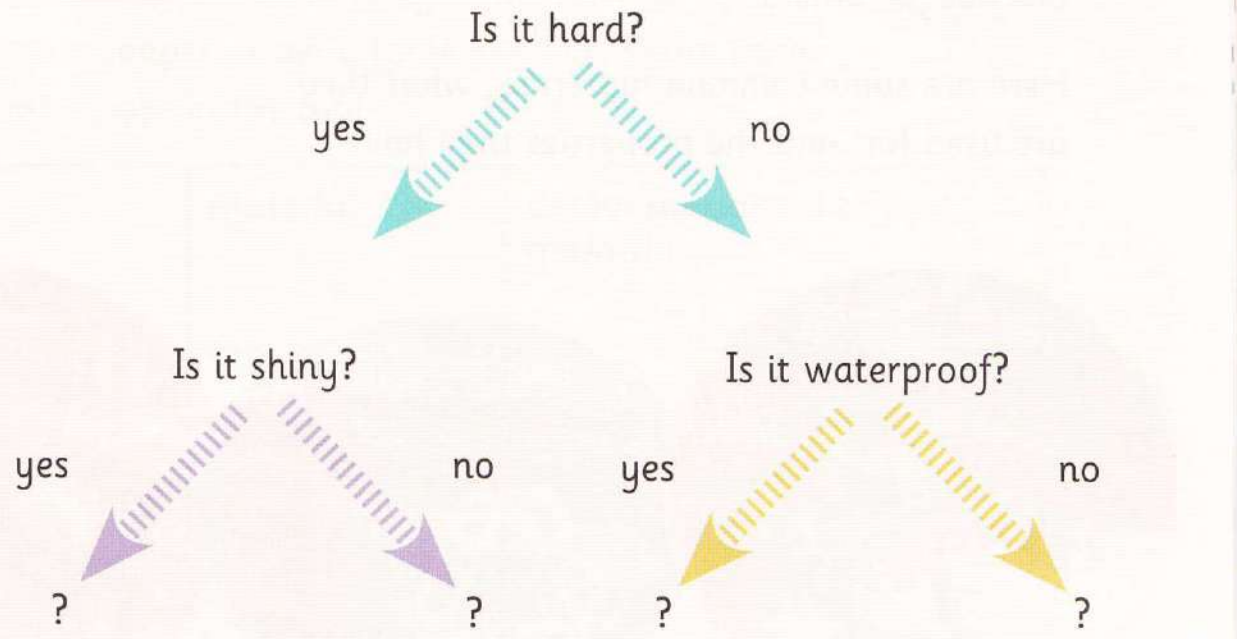
Look at the picture to see what to do.



Now choose another property and sort the objects into two different groups.

There are different ways of sorting. A branching tree database can be used to identify these four objects.
Choose one of the four objects and answer the questions.





Questions

1 Write **three** different materials that are waterproof.

2 Write all the properties you can think of for plastic.

Graphene is a new material that is very strong and very light.



Talk about it!

What could graphene be used for?

What you have learnt

- Materials can be sorted in different ways.
- A branching tree database can be used to identify objects.

5.3 Uses of materials

Every material has properties. The properties make a material good for making some objects but not for others.

Here are some common materials, what they are used for, and the properties they have.

Word to learn
observation





Wood is hard, strong and easy to make into objects such as chairs or bookcases.



Plastic is strong, waterproof, and easy to shape into objects such as bottles and bins.



Metal is strong, does not burn and can be sharp. Scissors, pans, knives, forks and spoons are all made from metal.

Sensible and silly materials

Metal is a sensible material for a pan because of its properties. What would happen if the pan was made of wood?

Here are some other objects made from silly materials.





Activity 5.3

You will need:
objects in the classroom that are made of only one material

Why is that material useful?

Find objects in the classroom that are made of only one material.
Talk about why each object is made of that material.
On a large sheet of paper, draw a table like this. Write each **observation** you make in the table.

Object	Material	Useful properties of that material
chair 	plastic	Strong so it does not break. Flexible to make it comfy. Light so it is easy to move.



A car made from ice!

Questions

1. Which material would be good for making sunglasses? Why?
2. A long time ago, knights used to wear metal armour. Why is metal not used for clothes?

Talk about it!
Why is a car made of ice a silly idea?

What you have learnt

- How material is used depends on its properties.

5.4 Testing materials

Activity 5.4a

You will need:
some fabrics • paper • metal foil • a small funnel
measuring cylinder • a timer

Which material is best for an umbrella?

An umbrella needs to be waterproof.

Predict which material will be the most waterproof. The waterproof materials will not let the water through.

Look at the pictures to see what to do.

Write your results in a table.

What will you keep the same to make this test fair?

Look closely at the materials that are not waterproof. Can you see why they are not waterproof?



1



Set up your equipment as shown.

2



Place your material in the funnel.

3



Pour water into the material and start timing.

4



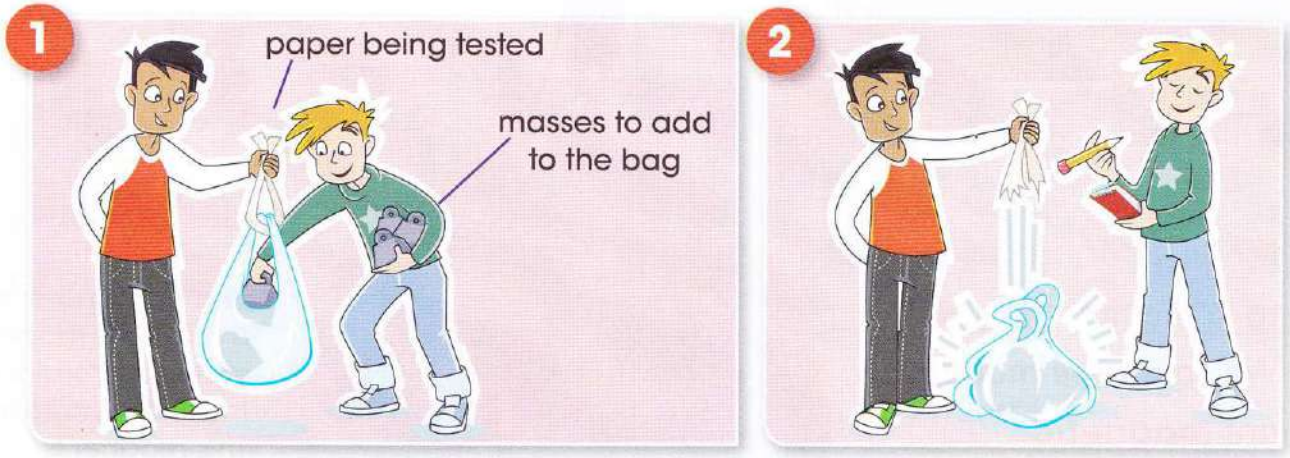
Is there any water going into the measuring cylinder?

Activity 5.4b

Which paper would make the strongest paper bag?

You will need:
thin pieces of different types of paper
some masses • a strong plastic bag

Look at the pictures to see what to do.
Predict which paper will be strongest.
How will you make this a fair test?



Write your results in a table.
Look closely at the strongest paper. Can you see why it is strong?

Questions

1 Why do the pieces of paper have to be the same size?
2 Is paper a good material for a bag? Why?

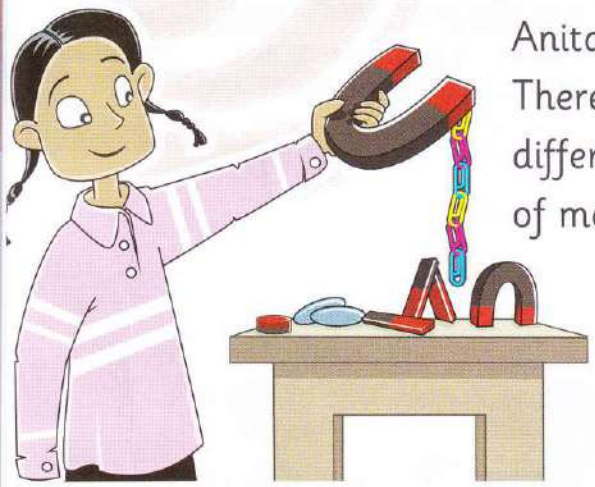
Talk about it!

How could you investigate which paper towel is the most absorbent?

What you have learnt

- Objects that need to be waterproof have to be made from waterproof materials.
- Objects that need to be strong have to be made of strong materials.

5.5 Magnetic materials



Anita has a **magnet**. There are lots of different types of magnets.

Words to learn

magnet attracted
magnetic non-magnetic
pattern



Magnets are **attracted** to some materials. We say these materials are **magnetic**. Materials that are not attracted to magnets are **non-magnetic**.



Magnets can be used to separate magnetic materials from non-magnetic materials. This magnet is separating magnetic metal from other rubbish for recycling.

Sunil is trying to put a fridge magnet on the cupboard door but it keeps falling off. Why?



Questions

1. Name a magnetic material.
2. List three materials that are not magnetic.
2. Why did Sunil's magnet not stay on the door?

Activity 5.5

Which materials are magnetic?

You will need:
a magnet • some materials to test

Help Sunil by testing some materials.
Look at the pictures to see what to do.







Predict which materials will be magnetic and then investigate.
Draw a table like this one for your results.

Object	Material	Magnetic	Non-magnetic
chair	plastic		✓

Look at your results. Can you see a **pattern**?
Are some materials always non-magnetic?
Which materials are magnetic?

Talk about it!
What else could we use magnets for?

What you have learnt

- Some materials are magnetic.
- Many materials are non-magnetic.

5 Check your progress

1 Write three properties for each of these materials.

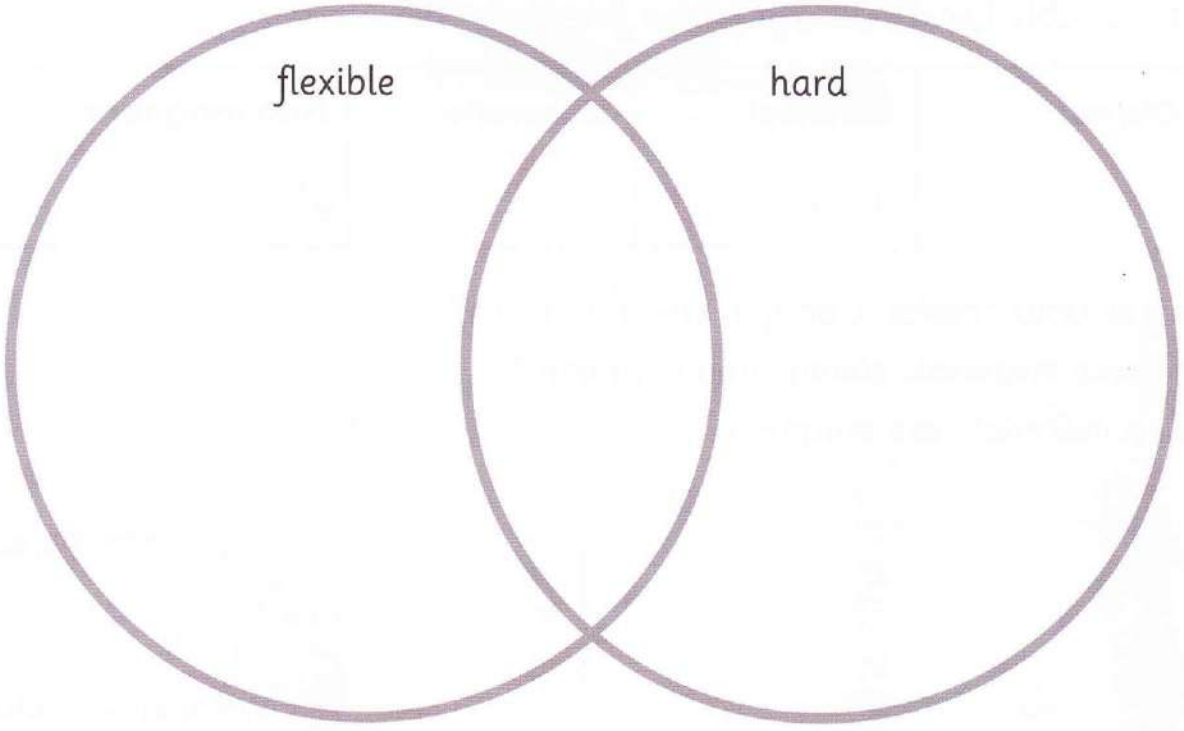
The first one has been done for you.

plastic | metal | glass | paper | stone

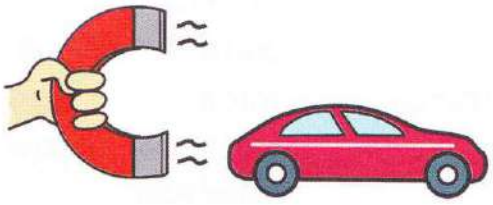
Plastic is smooth, flexible and light.

2 Copy this Venn diagram and write the following objects in the right places.

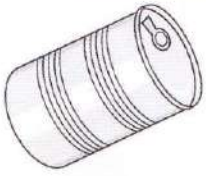
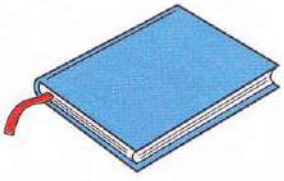
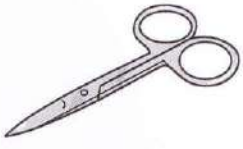
cotton T-shirt | wooden pencil | stone | metal paperclip | plastic ruler



3 Draw a picture to show what will happen to the metal toy car.



4 Copy and complete this table.

Object	Magnetic	Non-magnetic
a metal can 		
a book 		
a football 		
metal scissors 		

6 Forces and movement

6.1 Push and pull

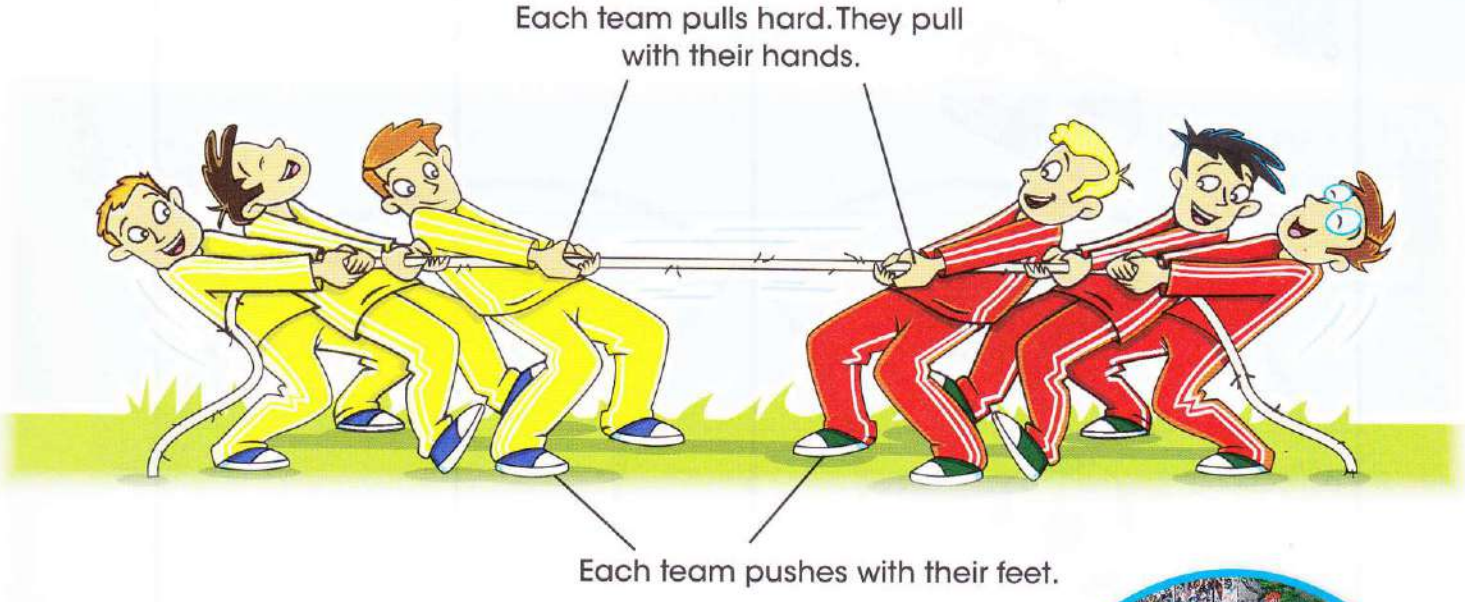
Every day you **push** and **pull** many things.
Push and pull are examples of a **force**.

Words to learn

push	pull
force	start
get faster	slow down
direction	



Six children divide into two teams to have a tug of war.
How are the children using pushing forces and pulling forces? What could they do so that they don't get hurt?



Forces can **start** and stop things moving.
They can also make things **get faster**, **slow down** or change **direction**.



A 'great tug of war' that involves thousands of people is held in Naha, Okinawa, Japan. It uses two ropes made of rice straw. Each rope is 2 m in diameter and over 300 m long.

Activity 6.1

You will need:
a ball • some water in a bowl
a balloon • a chair • a pencil

Forces start and stop things moving

Work with a partner and find ways to start each of your objects moving.
See if you can feel the push or pull you are using.
Now find ways to stop each of the objects from moving.
See if you can feel the force you are using this time.
Finally, find ways to change the direction in which each object moves.

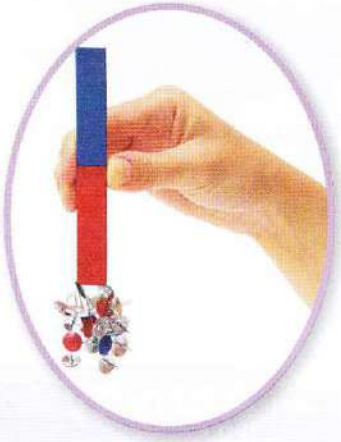


Challenge

How could you investigate how the size of the push or pull you give to a ball affects how far it moves?

Questions

1. Make a list of forces you have used so far today.
2. List **five** things you move with a pulling force.
3. Name a game in which you change the direction of a ball.



Magnets can be used to pick up magnetic objects.

Talk about it!
What have magnets got to do with forces?

What you have learnt

- Push and pull are examples of a force.
- We can use forces to start things moving, stop things moving and change their direction.

6.2 Changing shape

Forces can change the shape of things.

Words to learn
observe effect





A baker uses a force when making bread.



This potter is using a force to create a clay pot.



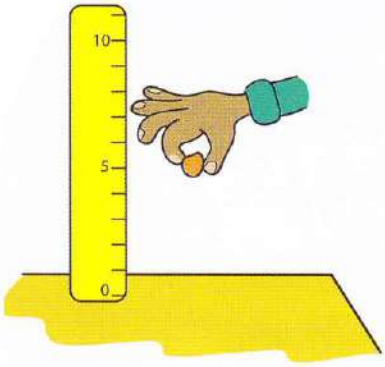
This carpenter is using a force to carve the wood.

Activity 6.2a

Investigating forces

You will need:
a ball of clay • ruler

Drop a ball of clay from a height of 5 cm.
Observe what happens to the ball of clay.
Write down your observations or draw what the clay looks like.
Shape the clay into a ball again. Drop the ball from a height of 10 cm. What is the **effect** on the ball of clay?
Drop the ball of clay from three more different heights.
Use ideas about forces to explain the pattern.



Activity 6.2b

You will need:
a small, heavy ball • different objects
a paper tube

Dropping a ball onto objects

Try using your hands to change the shape of the objects in front of you. Which were easy to change? Use a wide, tall paper tube to drop a heavy ball onto the objects.

First, predict what will happen to each material. Use a table like this.



Object	Prediction	Result
biscuit	break	broke into 5 pieces
clay		
raw vegetable		
cooked vegetable		
stone		

Plan and carry out this test. How will you make it a fair test? Record the results.

Questions

1. Do all materials change shape when they are pulled or pushed by hand?
2. List some workers who have to change the shape of materials.

What you have learnt

- Forces can change the shape of things.

Talk about it!
How do forces in nature change the shape of the land?

6.3 How big is that force?

There are small forces and big forces.



Omar pulls his toys with elastic. He is pulling with more force on some toys. How can you tell that this is the case? What toy do you think needs the biggest force?



This girl is using a small force to pull on the empty sledge.

It takes a bigger force to pull this vehicle.



A volcano has so much force it can blast large rocks high in the air.



Activity 6.3

Investigating how much force

You will need:
a balloon • objects to push against

Push against different objects with a balloon until they move. The bigger the force you need to move the object, the more squashed the balloon will be.

Choose **five** objects to push against. Record your results. Say whether you used a very small force, a medium force or a large force.



Questions

1 Put these in order from the biggest force to the smallest force:
- a horse pulling a cart
- a girl lifting a book
- a train pulling wagons
- a bird picking up a leaf.

2 Saida stretches an elastic band. Is this a pulling force or a pushing force?



All sorts of large things can be moved with a large force.

What you have learnt

- There are different sizes of force.

Talk about it!
How could you measure the size of a force?

6.4 Forcemeters



If you push on a door with foam you can see the size of the push.

Words to learn
forcemeter newton



Aleksy and Peng both push on a piece of foam. We can see how hard they push.

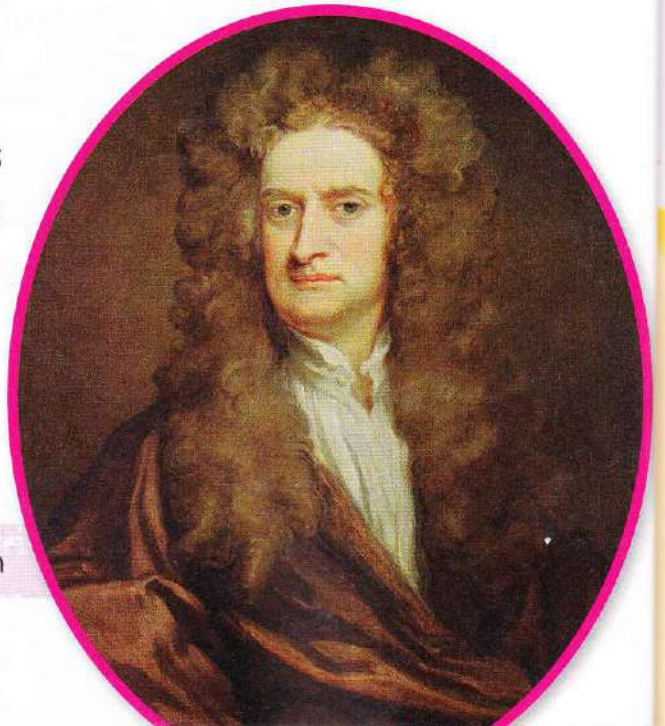




Forcemeters measure the size of pulls.

A **forcemeter** is a piece of equipment that measures the size of a force. Forcemeters measure the size of pulls.

The unit of force is the **newton** (N). The unit is named after the famous scientist Isaac Newton, who did a lot of work on forces.



Sir Isaac Newton

Activity 6.4

You will need:
a forcemeter

Measuring forces

Use a forcemeter to measure different pulling forces around your school. Before you measure, predict which object will need the biggest force to pull it. Record the sizes of the forces in a table. Draw a bar chart to show the results. Compare your predictions with what you found out.



Tennis players use forces of different sizes to control the ball.



The forces in the ropes will be different for different climbers.



A cycle helmet must be strong enough to protect the cyclist if their head gets hurt by a force.



The wind creates a force on the sail which moves the boat through the water.

Questions

1 What is the unit of measurement of force:
a a newton **b** a forcemeter **c** a pull?

2 Why is it important for equipment for sport to be strong?

Talk about it!
What could you use to measure pushing forces?

What you have learnt

- There is simple equipment which we can use to measure forces.
- The unit of measurement of force is the newton.

6.5 Friction

Friction is a force that acts when two surfaces rub together. Some materials cause more friction than others.

Words to learn

friction	grip
smooth	rough



Rub your hands together. Feel them rubbing and getting warm.



When hands are rubbed together friction acts between them.



You can go very fast down a slide. There will be friction between your clothes and the surface of the slide.

Friction can start and stop things moving. You need the friction between the floor and your shoes to help you start walking.



There is friction between the bottom of your shoes and the floor which helps your shoes to **grip** the floor surface. This stops you from sliding.

Ice is slippery. There is little friction between the puck and the ice and so the puck slides easily when it is hit by the player's stick.



The friction between the ball and the bat helps the ball to change direction.

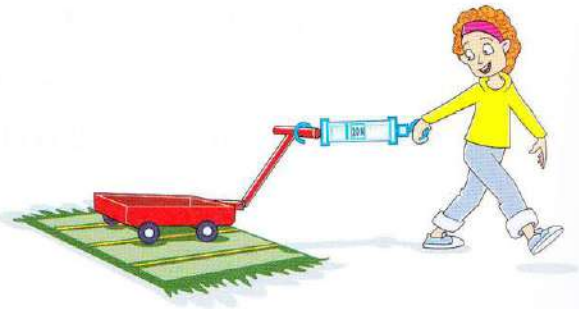
If you roll something across a surface, friction will cause it to slow down. It will stop at some point. Friction also help things to change direction.

Activity 6.5

You will need:
a small trolley • a forcemeter • different types of surface

Forces and friction

You can use a forcemeter to find out about friction. If there is more friction, you will need a bigger force to make something move.
Plan an investigation.
You will pull a trolley across different surfaces with a forcemeter.
Predict which surface will produce the most friction. Try using a very **smooth** surface and a **rough** surface.
How will you measure this? How will you record your results?
When you have done the test, decide what your conclusion is.
If you have time, test other surfaces.



Questions

1 When does friction act?

2 Faye loves her garden slide. She finds that when she wears cotton clothes she goes faster than when she wears woollen clothes. Explain why.

Talk about it!
Why are some shoes safer to wear on ice than others?

What you have learnt

- Friction acts when two surfaces rub together.
- The size of the friction force depends on the two surfaces which are rubbing together.

6 Check your progress

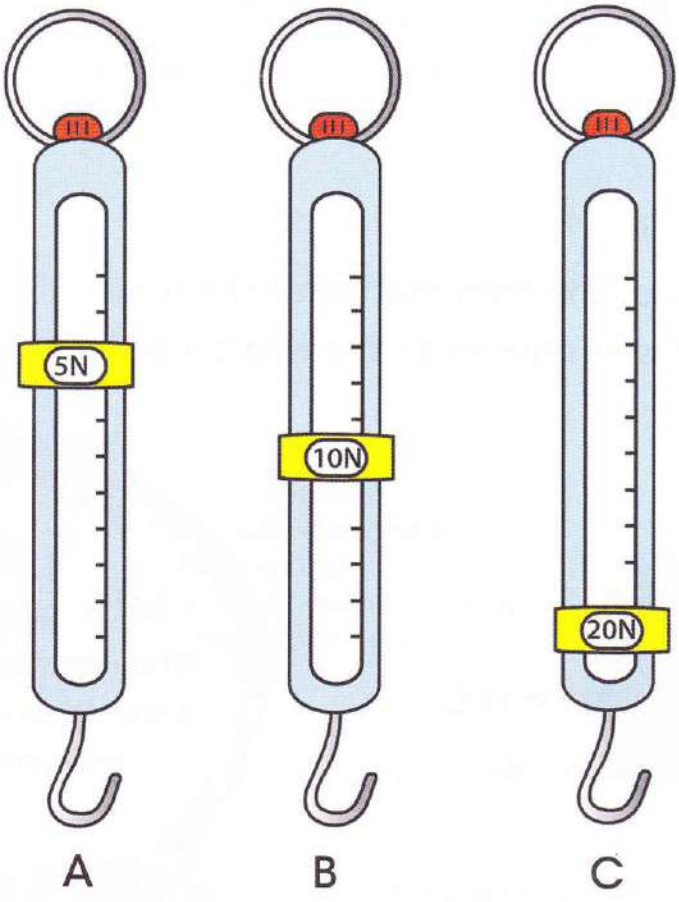
1 Here is a ball bouncing toward Kamili.



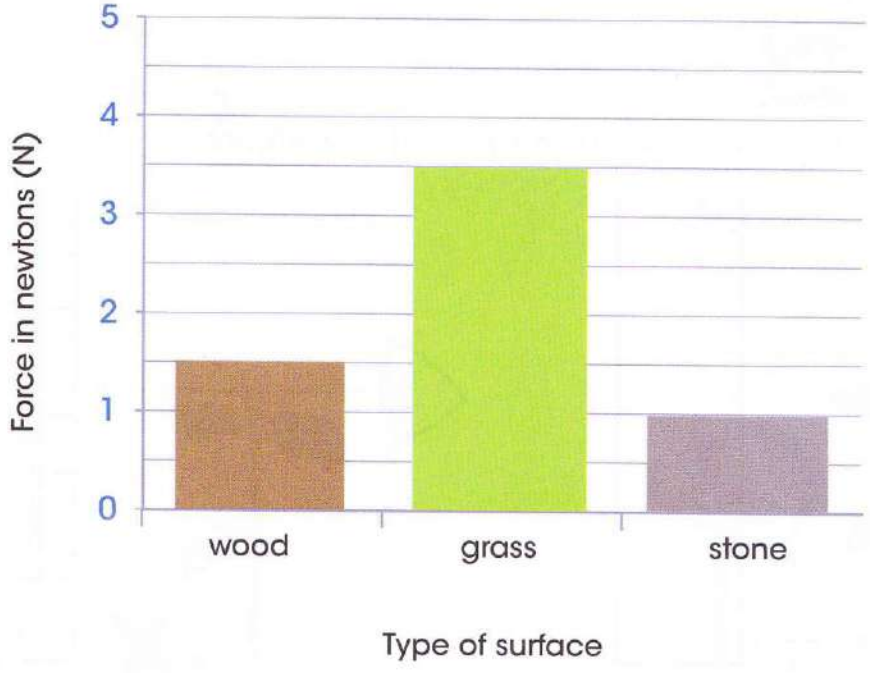
a How can she stop it?

b How can she change the ball's direction?

2 Which forcemeter is measuring the greatest force?



3 Alex tested the force needed to pull his skateboard over different surfaces. Here are the results.



a Which surface was the hardest to pull the skateboard on?

b Which surface was the easiest to pull the skateboard on?

4 Which surface creates the most friction with a skateboard:

- wood
- grass
- stone?

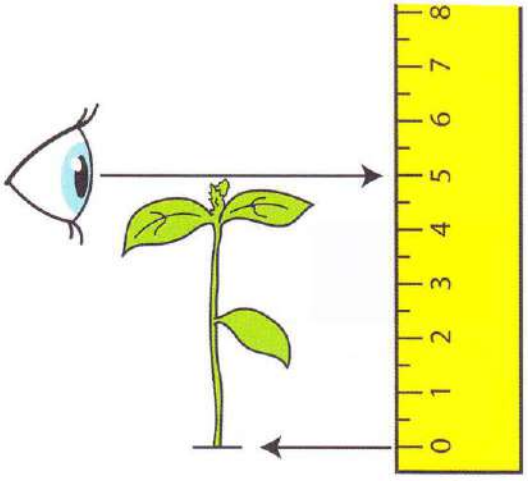
Explain your answer.

Reference

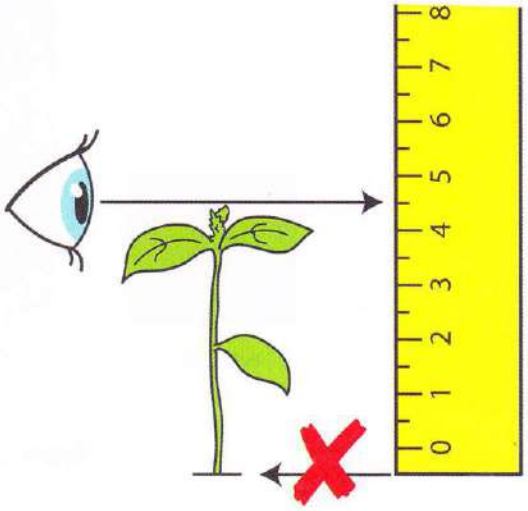
This section of the Learner's Book covers some of the new scientific enquiry skills for this stage. They build on the skills already gained from previous stages. You should refer to these skills whenever you need them.

How to use a ruler

Put the zero on the ruler next to the end of the object.

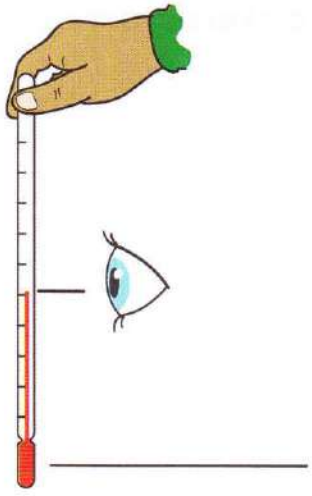


Put your eye level with the top of the object to read the scale.



Be careful. Often the zero is not at the end of the ruler.

How to use a thermometer to measure air temperature

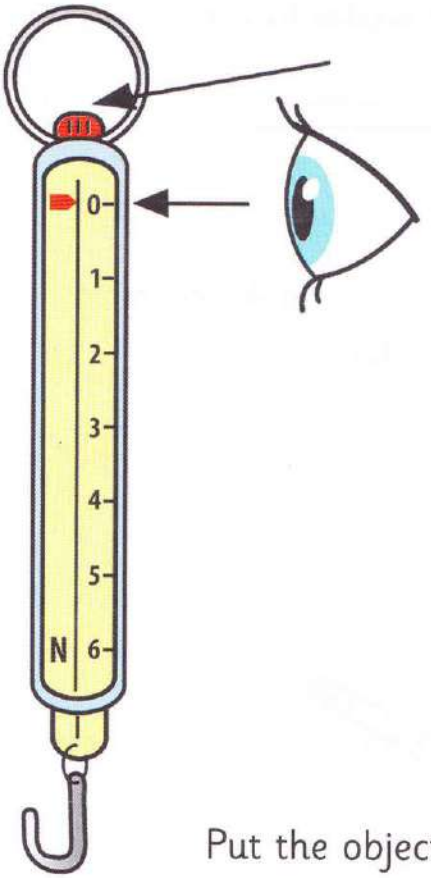


Hold the thermometer at the top.

Put your eye level with the top of the liquid to read the scale.

Do not hold the bulb or the thermometer will measure the temperature of your fingers.

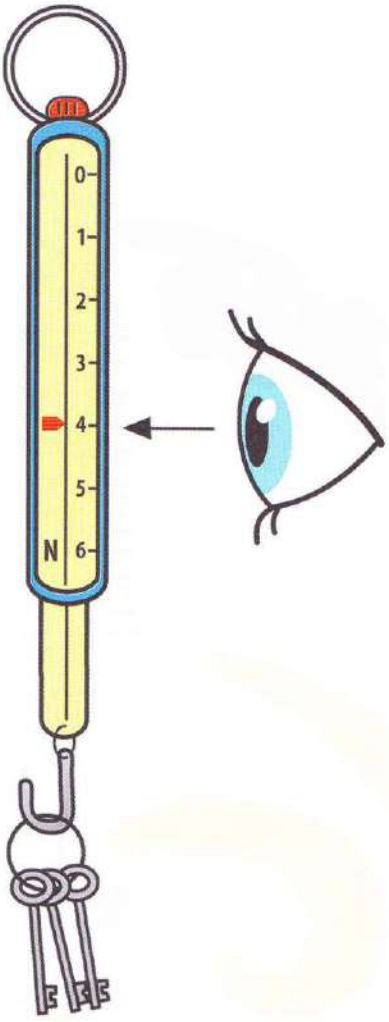
How to use a forcemeter



Turn the nut to check the forcemeter is set to zero.

Put the object on the forcemeter.

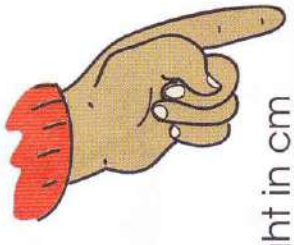
Put your eye level with the pointer and read the scale.

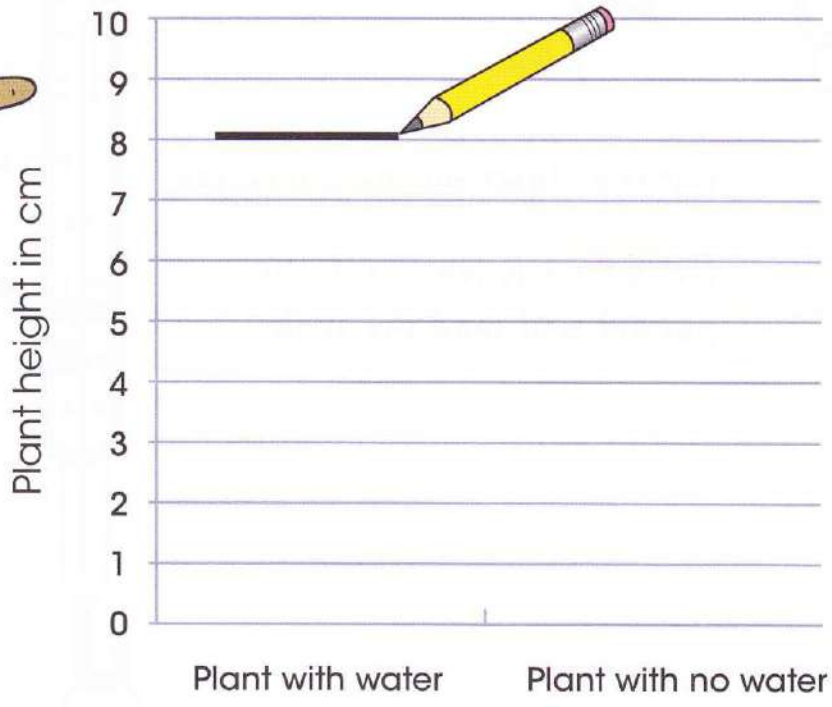


How to present results in bar charts

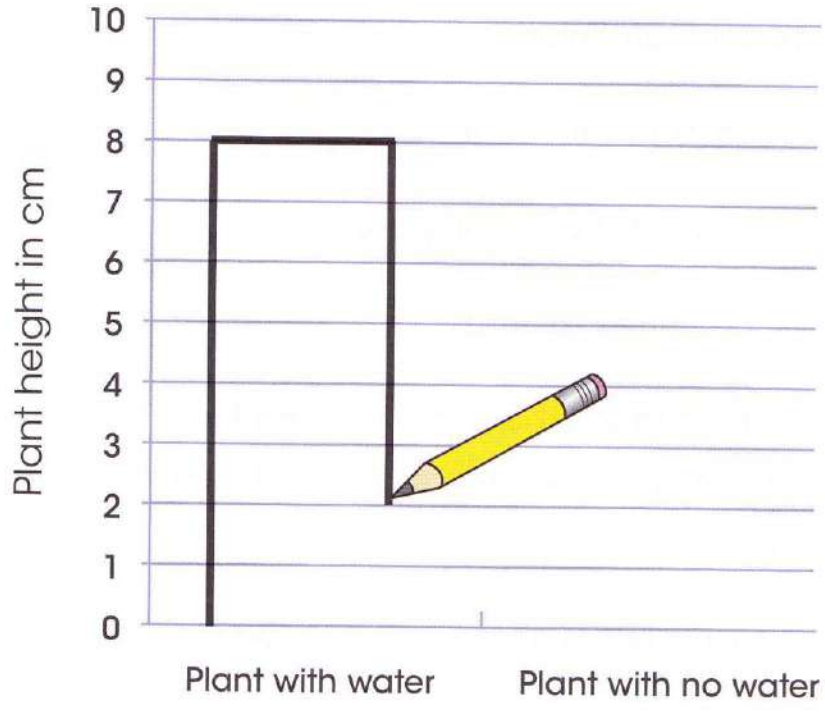
Plant height with water in cm	Plant height with no water in cm
8	4

Look at the table. The numbers tell you how tall to make each bar.
Draw a line at the top of the first bar. Use a ruler.

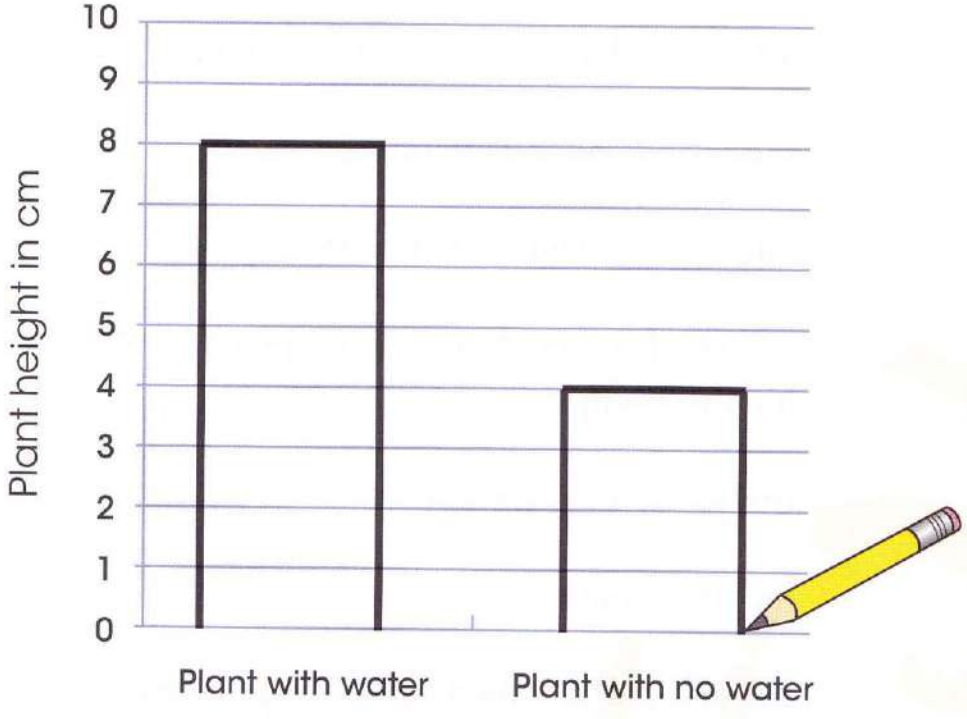




Draw the sides of the first bar. Use a ruler.



Draw the second bar the same way. Use a ruler.



Glossary and index

		Page
absorb	to soak up liquid	10
absorbent	a material that soaks up liquid	46
adult	an animal that is fully grown	28
attract(ed)	to be pulled towards	54
bar chart	a chart that shows results using bars (the length of each bar shows the size of each result)	12
bitter	a taste that is sharp and not sweet	40
breathe	to take air in and out of your lungs	27
carbohydrate	food that gives the human body energy	16
conclusion	what you find out in an investigation	12
dairy	foods that have milk in them	16
data	information that is numbers or facts	33
dehydrated	a lack of water that is damaging to the human body	19
diet	the foods that we eat	18
different	not the same	32
direction	the line along which something moves	58

discuss	to talk about and share ideas	21
dull	something that is not shiny	48
effect	the result of something happening	60
energy	what is needed by humans to do any action	16
excrete	the life process of getting rid of waste	27
exercise	moving around so that your heart beats faster	22
explain	give reasons for	9
eyesight	the sense that uses your eyes to see	42
fair test	controlling a test by only changing one thing and keeping other things the same	8
fat	food that the human body stores	16
feed	bring food	29
fingerprint	the lines on the tip of a finger	32
fish	a source of protein	16
flexible	can be bent	46
flowers	part of the plant where seeds are made	6

force	a push, a pull or a twist	58
forcemeter	something used to measure force	64
friction	the force between two objects when they rub together	66
fruit	part of a flowering plant; some fruits can be eaten	16
get faster	to move more quickly	58
grip	to hold on to a surface	66
group	put things with other things that have similar charecteristics	34
grow	to become larger and more developed	27
hard	not easy to squash, not soft	48
headache	pain that makes your head hurt	19
healthy	being well and free from disease	7
identify	to name	32
investigation	a test or experiment to find something out	8
leaves	part of the plant where the plant's food is produced	6
life processes	things that all living things do	27

magnet	an object that attracts magnetic materials	54
magnetic	a material that is attracted to a magnet	54
meat	a source of protein	16
newton (N)	the unit of force – force is measured in newtons	64
non-magnetic	a material that is not attracted to a magnet	54
nutrition	the life process of getting food for health and growth	18
observe	to look closely to find things out	60
observation	things that you notice when you look closely	51
offspring	the young of an animal	31
pattern	a link between results	55
predict	to think carefully about what might happen	8
property	what something is like, for example: a mirror is smooth and shiny	46
protein	food that the human body uses for growth and repair, for example: meat and fish	16
push	to use a force to move something away from you	58

pull	to use a force to move something towards you	58
question	a sentence that states what you would like to find out	35
record	to write or draw results to show what happened	20
reproduce	the life process of having babies, laying eggs or producing seeds	31
results	the observations or measurements made in a test	12
rigid	a rigid object keeps its shape, is not easy to bend or stretch, is not flexible	46
roots	part of plant that support the plant and collect water from the soil	6
rough	feels bumpy to touch	67
salt	small white crystals with a strong taste used in cooking	21
shiny	something that light bounces off	48
similar	when things are the same in some ways but not exactly the same	32
slow down	to move more slowly	58
smooth	something that is flat, not bumpy	67

soft	a soft object is easy to squash, not hard	48
sort	to put things into groups	27
sour	a taste, for example: vinegar or lemon juice	40
start	to begin to do something	58
stem	part of plant that transports water around the plant	6
sweet	the taste of sugar or honey	40
tally	a way of counting by drawing lines in groups of five	33
temperature	how hot or cold something is	12
thermometer	an object used to measure temperature	12
tongue	part of the body you use to taste things	40
transport	to move something	6
unhealthy	being unfit, unwell, sick or poorly	7
vegetables	the part of a plant that is grown to be eaten	16
waterproof	something that water cannot get through	46
wilt	to lose shape, become limp	8
young	not fully grown	27

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l = left, *r* = right, *t* = top, *b* = bottom, *c* = centre

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